

Collected and Discarded

Family Guarantees in U.S. Mortgage Underwriting

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The views expressed are those of the author and do not necessarily reflect those of the European Central Bank or the Eurosystem.

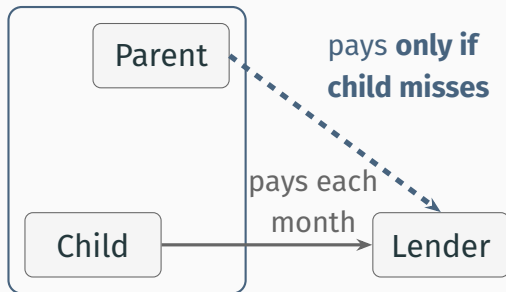
Young buyers need help, and a parent can cosign the mortgage

- Housing: main source of wealth
Kuhn, Schularick, Steins 20
- House prices outran incomes
Knoll et al. 17; Mabilie 23
- First-time buyers at a record low
National Association of Realtors 22
- More young buyers lean on family
Brandsaas 25
- Cosigning is rare in the U.S.

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both sign the mortgage



How does the lender value cosigning? The mortgage rules decide.

The parent counts for admission, not for the deposit or the price

Three rules govern about 65% of U.S. home mortgages as a result of the 2008 financial crisis:

Admission: is the loan made? (both must hold)

1. Income test (DTI)

payment $\leq$
share of income

Does the parent count?

✓ incomes pooled

2. Deposit test (LTV)

loan $\leq$ share
of house value

✗ ignored

Pricing: at what rate?

3. Price

rate from credit
score and leverage

✗ lower of two scores

Research question

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Do intergenerationally cosigned mortgages change the trade-off between **access to homeownership** and **financial stability** that mortgage rules strike?

► **Policy:** Should the deposit and the price recognise the cosigner, as the income test does?

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Why care

- **Stability:** rules date from the 2008 crash
- **Access:** 65% of under-35s rent (near record); 90% of young renters would like to buy but can't afford it

It does change the trade-off between access to homeownership and financial stability

Access fair price + deposit

Cosigning doubles, 4.8 → 9.7%

Owners +1.27 pp (+0.54 pp in GE)

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Missed payments rise

Foreclosures do not: parents pay

Preview of results

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The price ignores the guarantee

Fair price: a discount of 17 to 27 bp

Market charges +0.4 bp

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Welfare and distribution

+0.07%

more intergenerational inequality

Contribution to the literature

Housing and macroprudential regulation: Bolliger et al. 25; Castellanos et al. 24; Corbae–Quintin 15; DeFusco et al. 20; Elenev et al. 16; Greenwald 18; Hurst et al. 16; Jeske et al. 13; Kaplan et al. 20

⇒ Cosigning as measure to access mortgages without adding foreclosure risk

Asymmetric information and credit markets: Bester 85; Beyhaghi 22; Chiappori–Salanié 00; Einav–Finkelstein 11; Garcia-Villegas 26; Ioannidou et al. 22; Stiglitz–Weiss 81; Stroebel 16; Tzioumis 17

⇒ Third-party guarantee counted for admission, information discarded in pricing

Intergenerational transfers and housing: Allen et al. 26; Benetton et al. 22; Black et al. 22; Blickle–Brown 19; Boar 21; Brandsaas 25; Dalla Pellegrina–Scollo 16; De Nardi 04; Engelhardt–Mayer 98; Jakučionytė–Singh 22; Kaplan 12; Nekoei–Seim 23; Paz-Pardo 22

⇒ First U.S. measure of parental cosigning and transfer via insurance

Roadmap

- 1 Data**
- 2 Static model**
- 3 Natural experiment**
- 4 Quantification**
- 5 Counterfactual**
- 6 Conclusion**

Data

Measuring parental cosigning

I estimate 5.5% of mortgages are intergenerationally cosigned:

Loan records (HMDA)

10.5 million purchases, 2018–24

Parent = co-applicant **20+ years older**

Sees at most two borrowers

⇒ **2.5%** of purchases

County deeds, web-scraped

31 NY counties, 4.31 million loans

Name **every** signer

Add the parent beside a couple

⇒ **3.5%** of purchases

Also: credit scores (NSMO survey), late payments (Fannie Mae, 2.3 million loans)

▸ age-gap rule

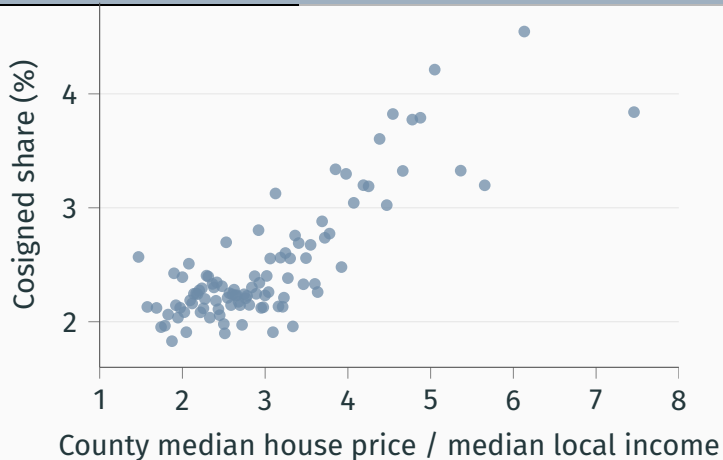
▸ web scraping

▸ deed sample

▸ deed signers

▸ sample validity

Fact 1: cosigning rises with price-to-income



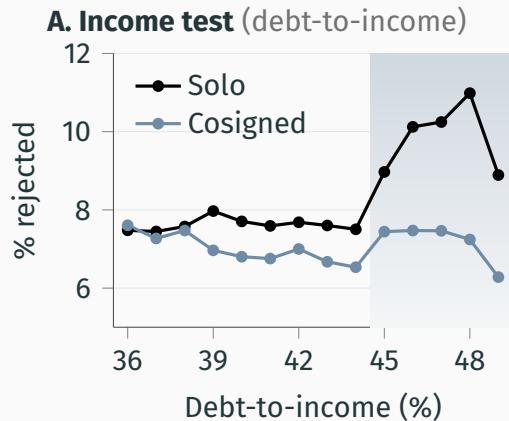
- 1,763 counties (12,278 county-years, 2018–24, pooled)

▸ 2024 only

▸ means

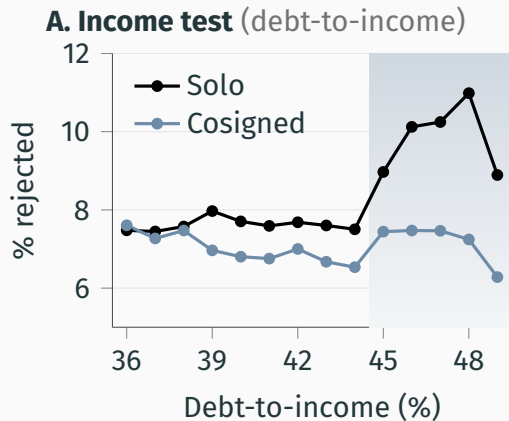
▸ controls

Fact 2: Rejection rates differ by admission test

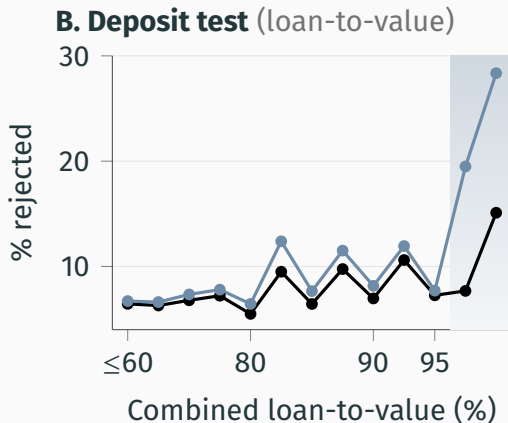


45-49: 7.1% vs. 9.8%

Fact 2: Rejection rates differ by admission test



45–49: 7.1% vs. 9.8%



At 96–97: 19.5% vs. 7.7%

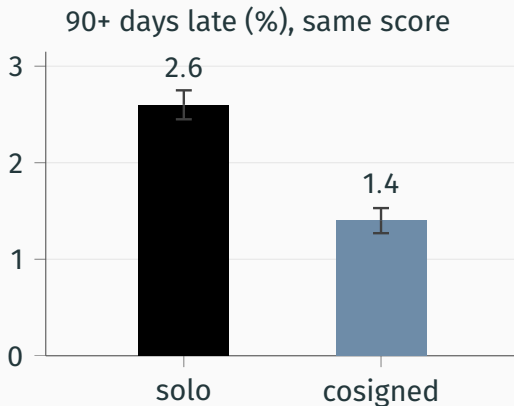
► who applies where ► full distribution ► controls ► why 45 and 50

Fact 3: Cosigned and Solo borrowers pay nearly the same rate



No discount for the guarantee: +0.08 pp inside a price cell

Fact 4: Cosigned loans default half as often



Parents cover missed payments before default

Static model

Static model: Information and Timing

Timing

1 **Parent** y_p , altruistic
knows y and the type
⇒ guarantee, or not

2 **Lender**
sees y , y_p , guarantee not type
⇒ a rate per contract

3 **Child** income y
own type: *safe* or *risky*
⇒ solo or cosigned only

1 **Parent:** guarantee, or not

Loan size, own assets and default are drawn; house size is a fixed ratio of the loan

Deposit cap is tighter when cosigned ($L^* = 95\%$ of P , against 97% solo); the income test runs the other way

▸ each actor ▸ admission tests ▸ pricing rule ▸ key assumption ▸ assumptions ▸ equilibrium ▸ three regions

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- 3 **Child**: solo or cosigned **pooled-income** test (Fact 2)
- 4 Then the child repays or **misses** parent pays what she can (Fact 4)

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Pricing: the rate reads the child, not the parent

Free entry, so each contract breaks even on what the **child** is expected to pay (ρ funding, g fee)

$$\mu_k(y) (1 + r_k(y)) = 1 + \rho + g \quad \implies \quad 1 + r_k(y) = \frac{1 + \rho + g}{\mu_k(y)}$$

► the rule, formally ► fair rate ► grid only ► how big

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But on a cosigned loan the lender is paid **twice over**

$$\left[\underbrace{\mu_{co}(y)}_{\text{the child pays}} + \underbrace{(1 - \mu_{co}(y)) \bar{p}(y_p)}_{\text{the parent covers}} \right] (1 + r_{co}(y)) L^* > (1 + \rho + g) L^*$$

Zero profit on *that* is the fair rate, and it is strictly lower

$$1 + \tilde{r}_{co}(y, y_p) = \frac{1 + \rho + g}{\mu_{co}(y) + (1 - \mu_{co}(y)) \bar{p}(y_p)} < 1 + r_{co}(y) \quad \text{whenever } \bar{p}(y_p) > 0$$

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The parent's payments are in the lender's **revenue**, never in the **price**

► the rule, formally ► fair rate ► grid only ► how big

The guarantee: Adverse Selection, Screen, Cure

Need (–)	+	Screen (+)	Cure (+)
a risky child values keeping the home more		a parent signs less for a risky child	the parent pays what the child misses
Net: <i>ambiguous</i> ► in the rate			Net: <i>positive</i> ► not in the rate

Admission adds both incomes, the interest rate is independent.

Selection buys access at higher default; the cure buys it with no more foreclosures.

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⇒ **Price:** fair cosigned rate **below** the rate charged ▶ formal

⇒ **Access:** no rate admits a buyer the deposit excludes ▶ formal

⇒ **Sorting:** richer safe children go solo, so cosigning reveals risk ▶ formal

▶ need and screen

▶ why ambiguous

Predictions: A cost shock to the lender

There are two margins of adjustment for a lender:

1. Admission responds differently

Only solo borrowers are turned away

Cosigned files cleared the income test on two incomes (Fact 2)

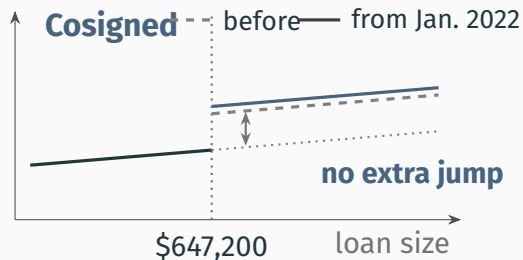
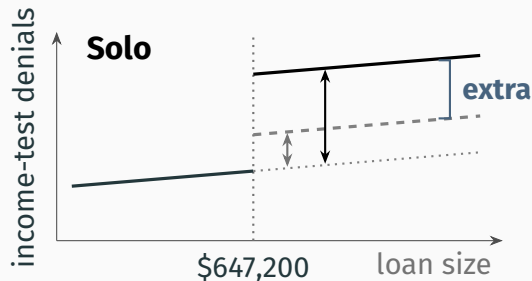
2. Rates respond similarly

The rate is set on the child's own risk in both pools (Fact 3)

A test the model was not built for → **2022 fee change**

Natural experiment

A 2022 fee change raises lenders' cost on larger loans only



Key assumptions: (1) composition at the limit constant over time (no pre-trends); (2) continuity at the cutoff in differences; (3) no switching into cosigning

► evidence

► the reform

► sample and design

► why not OLS?

► identification

What we estimate: how much the jump at the limit changed

$$Y_{ilt}^k = \alpha_{year}^k + \beta_1^k D_{it} + \beta_3^k (D_{it} \times Post_t) + f_{year}^k(\tilde{X}_{it}) D_{it} + Z_{it} \gamma^k + \delta_l^k + \epsilon_{ilt}^k$$

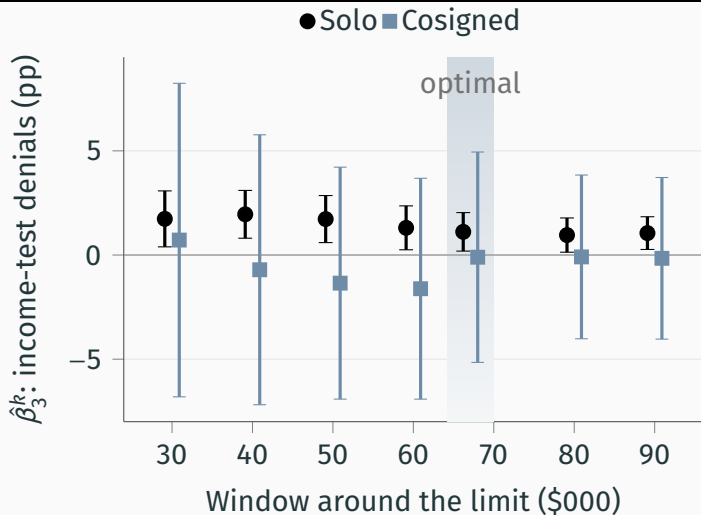
- $k \in \{\text{so}, \text{co}\}$: each pool estimated **separately**
- Y_{ilt}^k : income-test denial, loan i , lender l , year t
- $\tilde{X}_{it}$: running variable from threshold
- $D_{it} = \mathbb{1}\{\tilde{X}_{it} \geq 0\}$: above the limit
- α_{year}^k : year effects
- β_1^k : the jump at the limit **before** the reform
- β_3^k : **how much that jump changed**
 - ▶ the parameter of interest
- $f_{year}^k(\tilde{X}_{it}) D_{it}$: each year its own slope on each side
- $Z_{it} \gamma^k, \delta_l^k$: controls, lender effects

▶ sample and design

▶ why not OLS?

▶ identification

Lenders turned away solo borrowers; rates did not move



- $\hat{\beta}_3$: solo +1.11 **pp**,
cosigned -0.11 pp

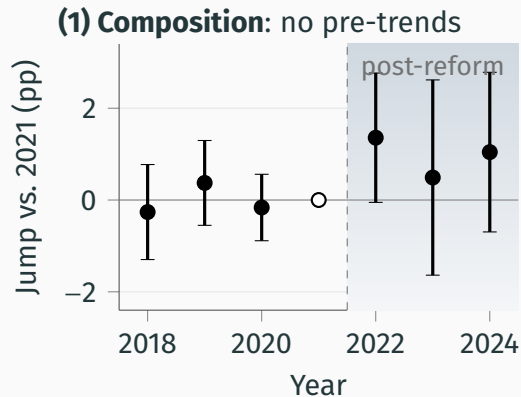
- Rates: no response
in either pool

► near the limit

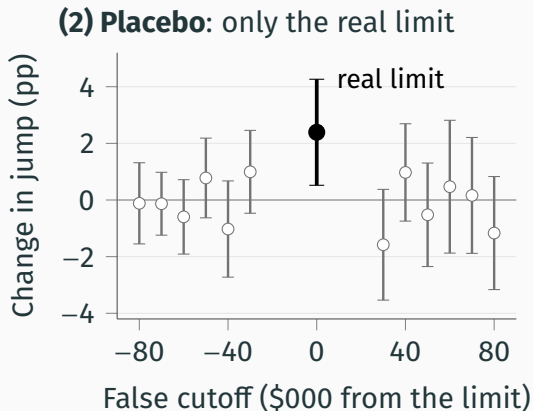
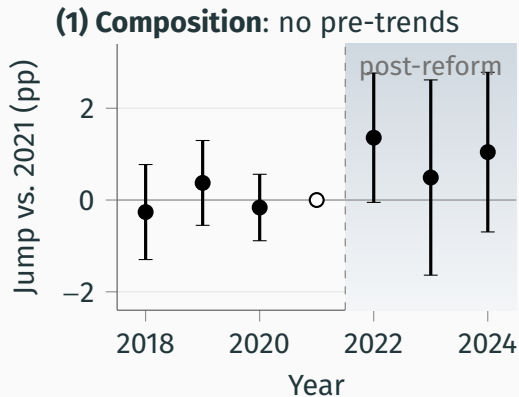
► denials table

► rates table

Validity: flat before 2022, and only at the real limit



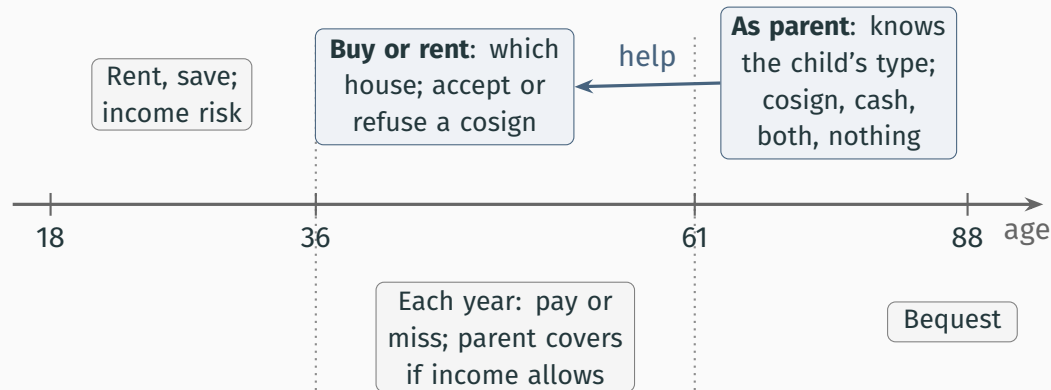
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► tables ► bunching ► 1. pre-trends ► 2. false cutoffs ► 3. switching ► identification

Quantification

A lifecycle model where every child later becomes a parent



- Hidden risk type; lenders break even pool by pool under today's mortgage rules

The child: take up the guarantee, then pay or default

1. Child at purchase rent or buy, which house, and whether to take the guarantee

$$V_{18}(\mathbf{s}) = \sigma_{rb} \ln(e^{V^{rent}/\sigma_{rb}} + e^{V^{buy}/\sigma_{rb}})$$

subject to $(1 - \lambda_k) p h \leq a$ and $m/Y_k \leq \tau_k$, with $Y_{co} = y_c + y_p$ **the reform moves** λ_{co}

2. Each year: pay or default an involuntary floor, then a choice

$$V_j^{PD} = \underbrace{(1 - \tilde{\pi}_\theta) V_j^{def}}_{\text{involuntary}} + \tilde{\pi}_\theta \sigma_{def} \ln(e^{V_j^{pay}/\sigma_{def}} + e^{V_j^{def}/\sigma_{def}})$$

solo default is foreclosure; cosigned default is **invocation**: the parent pays and the house is kept
model age 18 = age 36 on the timeline

► the problem ► preferences ► income ► penalties ► the cure

The parent screens, the lender prices the pool

3. Parent knows the child's type, chooses none / cosign / cash / both

$$V_{43}(\mathbf{s}, \theta) = \sigma_{\text{help}} \ln \sum_{\kappa=1}^4 \exp\left(\frac{u_{\kappa}}{\sigma_{\text{help}}}\right), \quad u_{\kappa} = \underbrace{V_{43}^p}_{\text{own value, carries the expected liability}} + \underbrace{\lambda_{\kappa}(z_p) \Delta V_c}_{\text{altruism} \times \text{child's gain}} + \underbrace{\eta_{\kappa}}_{\text{cost of helping}}$$

$$q_{\theta}(z_p) = \underbrace{\Pr(\text{offer} \mid \theta, z_p)}_{\text{parent}} \times \underbrace{\Pr(\text{accept} \mid \theta, z_p)}_{\text{child}}$$

4. Lender breaks even on the pool it believes it faces

$$\mu_k(h, z_c) = \tilde{\pi}_B + (\tilde{\pi}_G - \tilde{\pi}_B) q_G^k(h, z_c), \quad \Delta r = \frac{R(z_c)}{\mu_{co}} - \frac{R(z_c)}{\mu_{so}}$$

► **Negative** premium wherever the screen puts more Good types in the cosigned pool

Six moments targeted, five held out

Panel A. Targeted

Moment		Data	Model
M_1	Homeownership, 19–48 (%)	38.1	38.4
M_2	Default rate (%)	0.85	0.81
M_3	Cosigned share (%)	5.47	4.85
M_4	Cash-gift share (%)	25.0	26.2
M_5	Default ratio by score	6.9	6.2
M_6	Cosign–solo wedge (pp)	1.36	1.61

▸ estimates

▸ all parameters

▸ at a glance

▸ identification

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Panel B. Not targeted

Moment		Data	Model
U_1	Default wedge within cell (pp)	+0.56	+0.26
U_2	Rate premium (pp)	0.01–0.08	0.00
U_3	Income gradient cosigning	–0.051	–0.01
U_4	Parents prob. paying	0.74	0.95
U_5	Default wedge after cure (pp)	–1.00	–0.69

▸ estimates

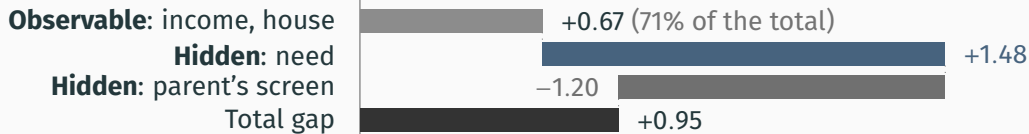
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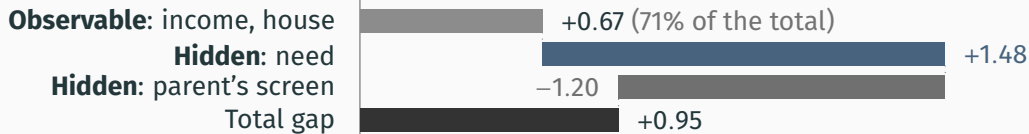
Result 1: Extra default risk is mostly observable

Default gap, cosigned – solo, in the model (pp)



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Default gap, cosigned – solo, in the model (pp)



Basis points a year, if the lender loses	25%	40%
Riskier pool	+35	+56
Parents pay when the child cannot	-52	-84
Kept by the lender	+18	+28

► table ► pp to bp ► fair premium

Counterfactual

Result 2: More owners, and no more foreclosures

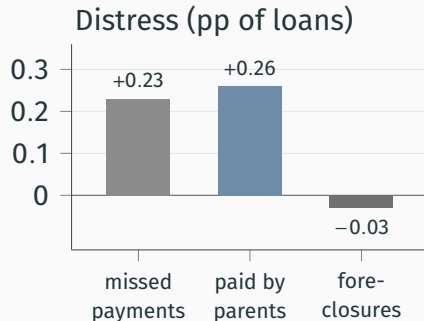
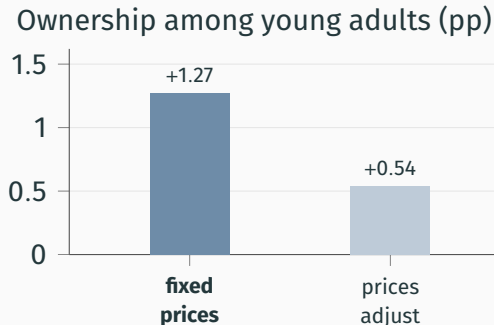
Reform: cosigned limit **90** → **97.5%** and **fair pricing**

Ownership among young adults (pp)



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▸ abroad

▸ supply

▸ if parents pay less

▸ why the deposit

▸ the new pool

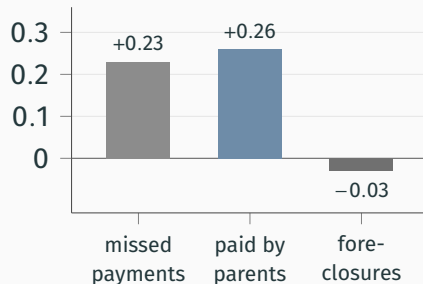
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Distress (pp of loans)



Cosigning doubles, 4.8 → 9.7%; cosigned rate **26 bp below** solo.

▸ abroad

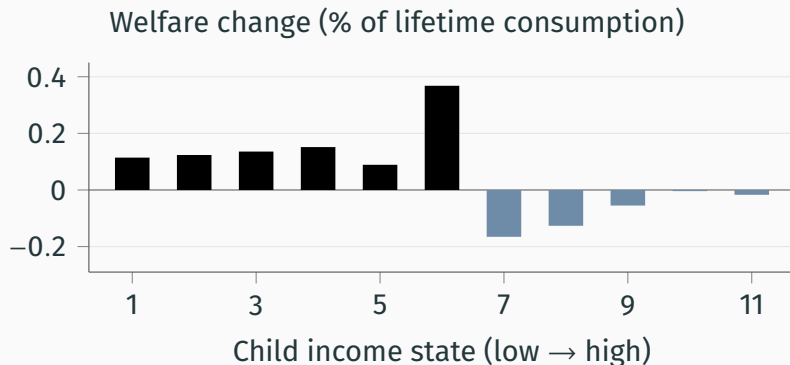
▸ supply

▸ if parents pay less

▸ why the deposit

▸ the new pool

Welfare: the lower half gains, the upper half pays

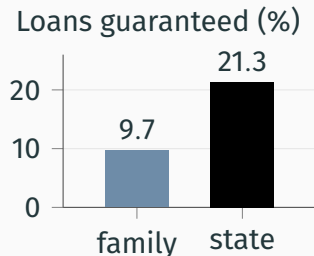


- Aggregate +0.074%
- Mean wealth +1.62%

- Bequests +2.50%
- Top-10% share -0.47

Horserace: The state guarantee

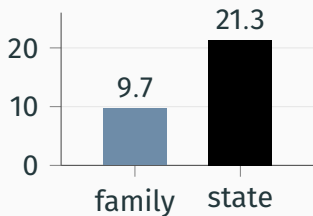
Ownership gap by parent income: **0.39** → **0.88 pp.** A state guarantee removes that line:



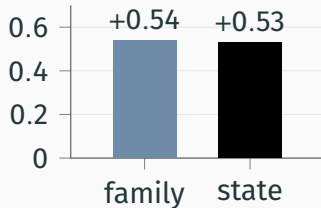
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Loans guaranteed (%)

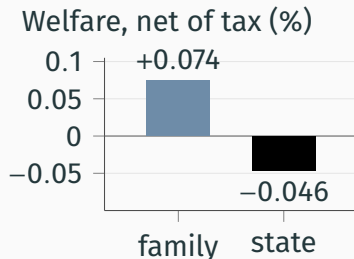
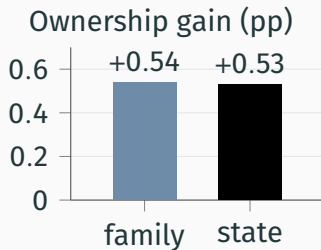
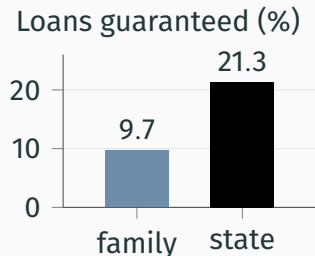


Ownership gain (pp)



Horserace: The state guarantee

Ownership gap by parent income: **0.39** → **0.88 pp**. A state guarantee removes that line:



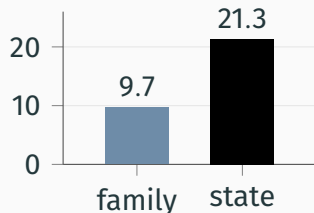
► who has a parent

► table

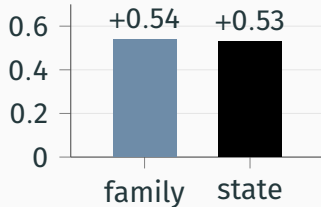
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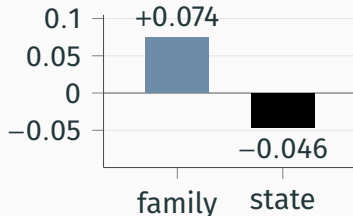
Loans guaranteed (%)



Ownership gain (pp)



Welfare, net of tax (%)



The state reaches everyone; a parent signs only *for this child*. Same pool either way: safe-type share **0.662**.

► who has a parent

► table

Conclusion

The answer, and what the paper still needs

Answer: intergenerational cosigning **changes the trade-off** between access to homeownership and financial stability that the rules strike.

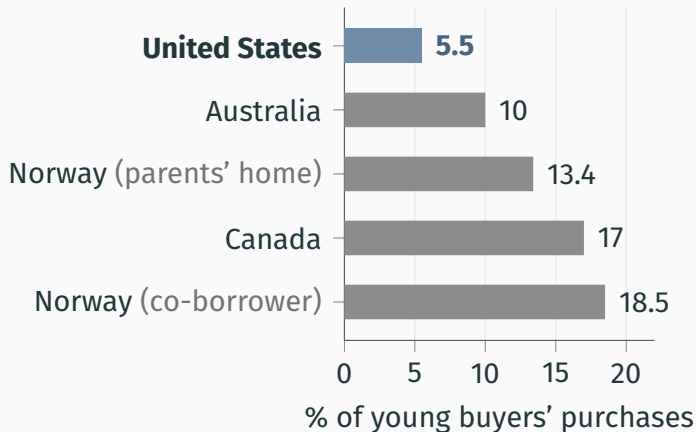
What the paper shows

- **Access without foreclosures:** owners +1.27 pp, foreclosures -0.03
- **The price is wrong:** fair is **17–27 bp below** solo, the market charges +0.4
- **The state cannot copy it:** no per-child screen ► the same ownership

Still missing

- Parents cannot default: absorption is an upper bound
- One purchase window: owning by the mid-thirties, not for life
- Prices are stationary: no price path, no price risk
- Parent income does not tilt the pool as the static model predicts

Backup: parental cosigning is rare in the U.S. and common abroad



- Abroad the guarantee often counts toward **deposit**
- There: 2 to 3 times as common
- U.S.: buyers under 45

▸ by country

◀ back

Backup: guarantees for young buyers, by country

Country	Share	What the guarantee reaches
A. Family guarantees		
United States	5.5%	Income test only: priced on the child, deposit unmoved
United Kingdom	7%	Income test only
Australia	10%	Deposit: guarantor's home lowers the loan-to-value
Norway, parents' home	13.4%	Deposit: lien counts toward the loan-to-value cap
Canada	17%	Co-borrowing; high-leverage risk insured nationally
Norway, co-borrower	18.5%	Income test only (debt $\leq$ 5 $\times$ income binds)
B. State guarantees		
Spain	15%	ICO: up to 20% of the price
Finland	19%	ASP: state guarantee on first-home loans
Italy	20%	Consap: the tranche above 80% loan-to-value
Portugal	24%	State-backed, with parents: up to 100% financing

Backup Orders of magnitude only. U.S.: model target (range 2.8–5.8%). Spain: eligible buyers; Finland: loan volume.

Backup: the age-gap rule finds parents

Rule (HMDA reports ages in bins)

- Buyer under 25: co-applicant 45+
- Buyer 25–34: co-applicant 55+
- Buyer 35–44: co-applicant 65+

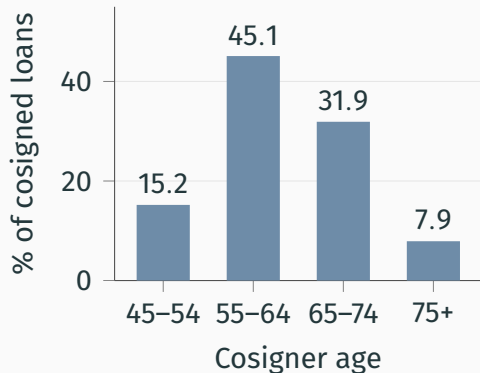
Who signs: father **58%**, mother 42%

280,649 pairs with both sexes reported

Survey check: non-occupant
co-signer **4.2%**

between age-gap 3.5% and either-rule
6.5%

NSMO, 8,249 first-time buyers; deed sample



Backup: how the deeds are scraped and matched

1. Scrape

County clerk
portals, 2018–24
Automated
browser: filter to
mortgages, month
by month
Open every
record: borrowers,
lender, town,
amount

2. Clean

Drop firms, trusts,
corrections, loans
< \$20k
Merge name
aliases
**Flag: more than
two borrowers**

3. Lender

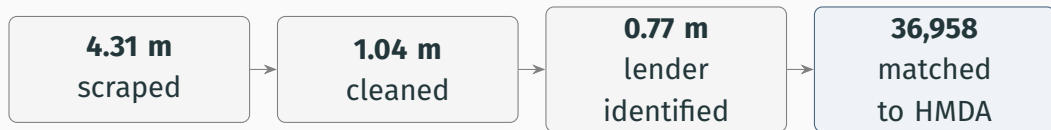
Strip MERS
nominee
Standardise bank
names
Match to global
legal entity ID
(GLEIF), as in
HMDA

4. Match

To HMDA loans
with:
Same year
Same town
Same lender ID
Same \$10k loan
bracket
Keep unique
matches only

◀ back

Backup: the deed sample in numbers



Deed rule (>2 signers)	1,286
Age-gap rule	1,304
Both	178
Either	2,412 (6.5%)

Matched: **16%** of eligible HMDA loans in the same counties and years from 2% (Staten Island) to 56% (Onondaga); NYC boroughs lowest

[◀ back](#)

Backup: is the extra signer on the deed a parent?

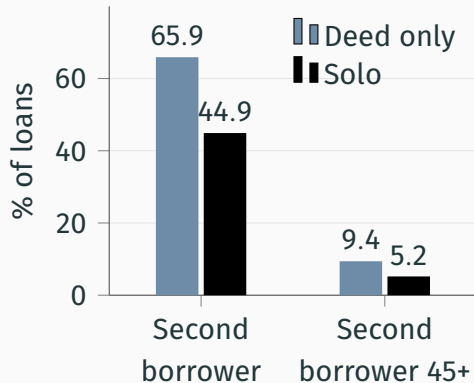
Noise checks

- ✓ Aliases as extra rows: **0.7%** of loans
- ✓ Firms, trusts: **2.4%** of flagged
- ✗ Surnames uninformative ($p = 0.40$)

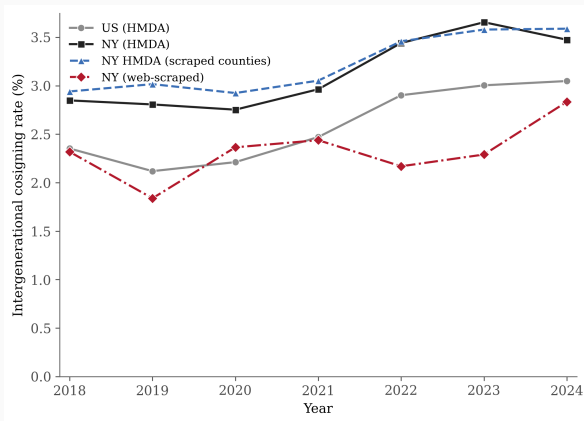
Positive test: deed-only vs. solo
with county + year FE

- Second borrower: +18.7 pp (s.e. 2.4)
- Second borrower 45+: +3.7 pp
($p = 0.03$)

One-sided test: with three signers and two HMDA slots, the parent can be missing from HMDA



Backup: the scraped counties look like New York



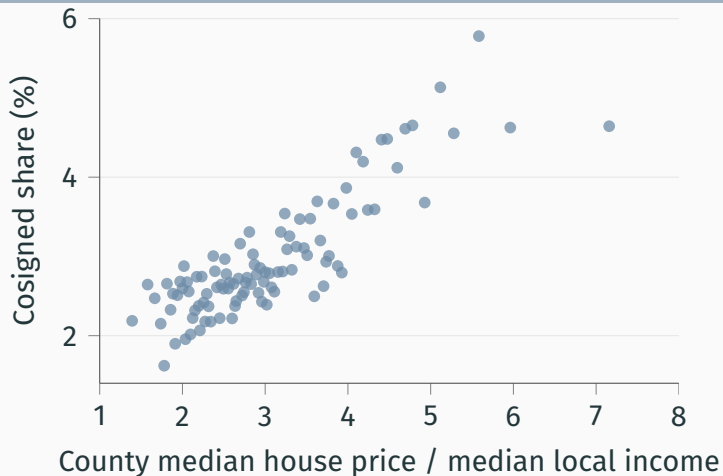
Scraped counties (**blue**) $\approx$ all of New York (**black**), which moves with the U.S.

(grey)

Backup

◀ back

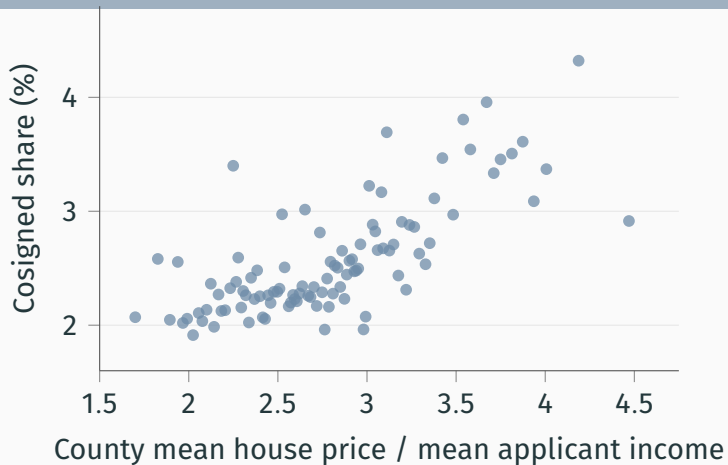
Backup: Fact 1 in 2024 alone



- 1,755 counties, 2024 only
- 100 equal-size bins
- Holds every year; **steeper since 2022**

◀ back

Backup: Fact 1 with means



- 1,763 counties, 2018–24 pooled
- Income: mortgage applicants (no mean in the Census data)
- Same gradient: cheapest fifth 2.3%, dearest 3.3%

◀ back

Backup: Fact 1 is not a big-city effect

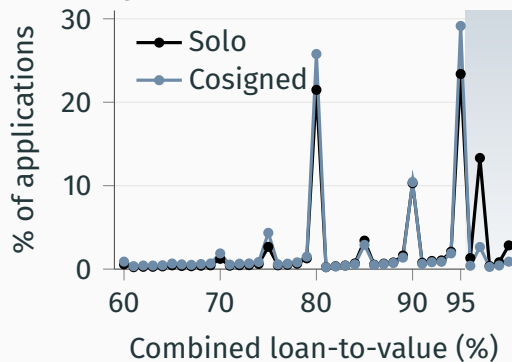
	Slope	(s.e.)
Raw	+0.43	(0.04)
+ income level	+0.45	(0.07)
+ density, metro, size	+0.40	(0.07)
+ buyer age, price growth	+0.40	(0.07)
+ state fixed effects	+0.15	(0.07)

Cosigned share (pp) on price / local income; 1,740 counties, loan-weighted.
Denser counties cosign *less* (-0.13); richer counties *less* (-1.34 per log income).

◀ back

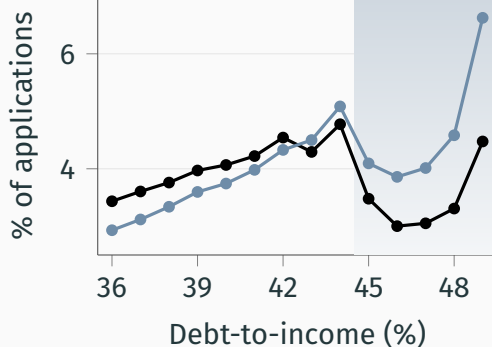
Backup: who applies where, on each margin

A. Deposit test (loan-to-value)



At 95: 29% vs. 23%; at 97: 2.7% vs. 13.3%

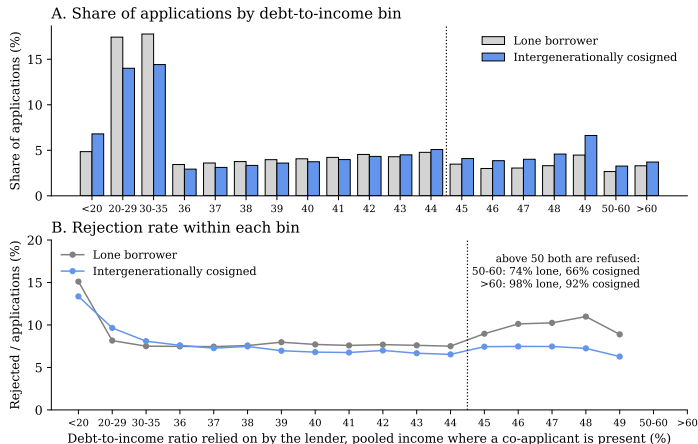
B. Income test (debt-to-income)



45-49: 23% of cosigned vs. 17% of solo

[◀ back](#)

Backup: Fact 2, the full debt-to-income distribution



◀ back

Backup: Fact 2 with controls

Dependent variable: application rejected (pp); DTI 36–49

	(1)	(2)	(3)
DTI 46–49 (vs. 43–45)	+0.66*** (0.17)	+0.46*** (0.17)	+0.68*** (0.17)
Cosigned × DTI 46–49	−0.64** (0.32)	−0.67** (0.32)	−0.88** (0.41)
Year, county, lender FE		✓	✓
Borrower controls		✓	✓
Years	2018–24	2018–24	2018–21
Observations	582,539	582,539	393,424
Lender clusters	1,663	1,663	1,483
R ²	0.001	0.046	0.051

Linear probability model; s.e. clustered by lender. *** $p < 0.01$, ** $p < 0.05$.

◀ back

Backup: why 45 and 50

Fannie Mae DTI limits

- Automated underwriting: up to **50%**
- Manual: 36%, up to **45%** with compensating factors
- Above 45, approval needs a stronger file

Where the parent enters

- Ratio on **pooled** income: the parent moves the file down the axis
- Automated system weighs risk across **all** borrowers
- Manual only: occupant's income alone, cap **43%**

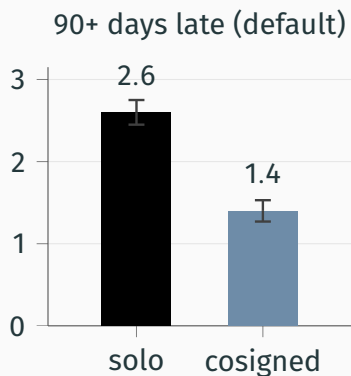
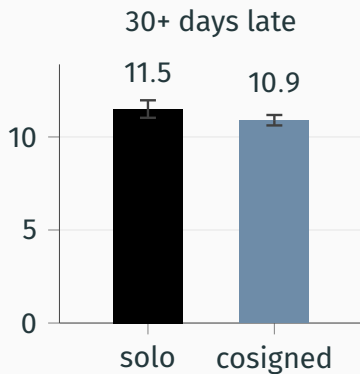
The parent acts as a compensating factor near the cap

Fannie Mae Selling Guide B3-6-02, B2-2-04

◀ back

Backup: Fact 4, 30+ and 90+ days late

Share of loans ever late (%), same credit score



Small gap at 30 days (−0.7 pp), large at 90 (−1.2 pp, about half the solo rate) [◀ back](#)

Backup: Facts 3 and 4, regressions

	Fact 3	Fact 4			
Cosigned – solo (pp)	Note rate	30+ late	90+ late	Forced exit	Recovered
No controls	+ 0.23 *** (0.02)	+2.32*** (0.20)	−0.31*** (0.07)	−1.48*** (0.15)	+6.27*** (0.58)
Same credit score	+0.22*** (0.02)	−0.68*** (0.14)	− 1.23 *** (0.07)	−1.68*** (0.16)	+9.34*** (0.56)
+ leverage	+0.25*** (0.01)	−0.36** (0.15)	−1.00*** (0.07)	−1.32*** (0.14)	+7.47*** (0.54)
Same price cell	+ 0.08 *** (0.01)	−0.24 (0.15)	−1.00*** (0.06)	−1.36*** (0.14)	+7.58*** (0.54)
+ DTI, state, loan size	+0.08*** (0.01)	−0.35** (0.16)	−1.03*** (0.06)	−1.28*** (0.14)	+7.83*** (0.50)
Solo mean	4.64%	11.5%	2.6%	2.4%	66.4%
Observations	2,295,805	2,295,805	2,295,805	234,299	176,191

Price cell: quarter × score × leverage. Last two columns: loans ever 30+ late. State-clustered s.e.

Backup

◀ back

Backup: linking Fannie Mae loans to HMDA

Fannie Mae performance files

Monthly status: late, forbearance, exit

Both credit scores, note rate

✗ no borrower ages

HMDA

Ages of applicant and co-applicant

✓ parent-child flag (20+ years older)

✗ nothing after origination

Match on year, state, county, **exact note rate**, loan amount band, term, leverage, DTI band, co-borrower

Keep a pair only if it is **unique both ways**

- **4.56 of 6.85 million** loans matched
67%; 55–84% by year
- Sample: **2.30 million** first-time buyers, 71,819 cosigned

- False matches: **0.3–1.4%**
placebo: every rate shifted by $\frac{1}{8}$ point
- Forbearance months count neither as late nor as exposure

Backup: the three actors, formally

Child chooses the contract (cosigned only if offered); if cosigned, a miss does not cost the home

$$\max_{k \in \{so, co\}} V_k(y, \theta) = \pi_\theta u(y - (1 + r_k)L_k, h_k) + (1 - \pi_\theta) \underline{u}_k, \quad \underline{u}_{co} > \underline{u}_{so}, \quad \pi_G > \pi_B$$

Parent knows θ and offers, $o = 1$, if her net gain is positive ($\eta_p \sim F_\eta$: taste shock)

$$\mathcal{U}_p(1) - \mathcal{U}_p(0) = \underbrace{\beta (V_{co} - V_{so})}_{\text{altruism}} - \underbrace{(1 - \pi_\theta) [u(y_p) - u(y_p - \text{payment she covers})]}_{\text{expected liability}} - \eta_p$$

Lender does not see θ ; free entry sets a break-even rate per contract

$$1 + r_k(y) = \frac{1 + \rho + g}{\mu_k(y)}, \quad \mu_k(y) = \mathbb{E}[\pi_\theta \mid y, k] : \text{the child's own repayment}$$

Backup: all assumptions of the static model

Assumption	Used for
Family knows the type, lender does not; prior $\lambda(y)$ rises with income	selection; screen read through y
Free entry; rate reads y and the contract, not y_p or the cover $\bar{p}$	the unpriced guarantee
House price and cosigned loan cap L^* fixed	rich children outgrow the contract
Key: $\Delta u_p > \beta (\Delta \underline{u} - G)$, liability outweighs the extra altruism	the screen, $q_G > q_B$
(A1) u concave, $u_{ch} > 0$, housing valued enough	gain from cosigning falls with income
(A2) Cosigning pays at low income, not at high income	interior cutoffs
(A3) Taste shocks: support $\mathbb{R}$, dispersed, log-concave	$q_B > 0$; unique pooling rate
(A4) Parent's liability bounded below: $\bar{p} > 0$, bounded y_p	no "bad news" at the top
Risky children still cosign at the separating rate	middle region (else two regions)

► existence, uniqueness

◄ back

Backup: from the static to the lifecycle model

Static model

Two types, repayment given

Prior rises with income

Parent offers or not

Taste shock η_p

Key assumption gives $q_G > q_B$

Pool quality by Bayes' rule

Rate on the child's own risk

Income test on pooled income

Three regions by income

One period, savings given

Lifecycle model

Same types; default is a **choice**, with a type-specific penalty

Prior rises with the child's productivity

Parent: cosign, cash, both or nothing; child: takes it up or not

Help cost plus extreme-value taste shocks

Not assumed: type-specific expected liability, $q_G > q_B$ endogenous

Realised share of safe buyers in each house-size and income cell

Observed rate schedule; cosigned gap by the same rule, cell by cell

Payment-to-income cap 40% solo, 42% on joint income

The trap: cells with a positive premium

Saving, buying, default chosen; children become parents; prices clear

Backup: equilibrium and the fixed point

A perfect Bayesian equilibrium: two cutoffs in the child's income y , rates $r_k(y)$, beliefs $\mu_k(y)$ with

1. **Parents** offer optimally at those rates: $q_\theta = F_\eta(\beta \Delta V - (1 - \pi_\theta) \Delta u_p)$
2. **Children** take the contract with the higher V_k
3. **Beliefs** follow Bayes' rule on income and the contract taken, $\mu_k(y) = \mathbb{E}[\pi_\theta \mid y, k]$
4. **Rates** break even per contract, $1 + r_k(y) = (1 + \rho + g) / \mu_k(y)$

Fixed point: rate $\blacktriangleright$ offers $\blacktriangleright$ pool $\blacktriangleright$ belief $\blacktriangleright$ rate. Where both types cosign,

$$r_{co}^{pool}(y) = T_y(r_{co}^{pool}(y)), \quad T_y(r) \equiv \frac{1 + \rho + g}{\pi_B + \Delta\pi \Lambda_1(y; r)} - 1$$

$\Lambda_1(y; r)$: share of safe children among the cosigned, given who offers at rate r

[▶ existence, uniqueness](#)

[▶ proof](#)

[▶ grid only: no loop](#)

[◀ back](#)

Backup: existence and uniqueness

The pooling rate (fixed point of T_y)

- Exists: T_y maps a closed interval of rates into itself, continuously
- Unique: $|T'_y| \leq C \sup f_\eta < 1$ if taste shocks are dispersed ► contraction

The two cutoffs (exit of safe, then of risky children)

- Exist: cosigning pays at low income and not at high income
- Unique: the gain ΔV falls with income at its zero; risky children cross later

Beliefs: Bayes' rule on the path; off the path, the worst case π_B

Scope: unique given the pooling schedule; static model only

Backup: proof, the pooling rate exists and is unique

$$T_y(r) = \frac{1 + \rho + g}{\pi_B + \Delta\pi \Lambda_1(y; r)} - 1 \quad \text{on } \mathcal{R} = \text{rates between pricing at } \pi_G \text{ and at } \pi_B$$

1. **Self-map:** the belief lies in $[\pi_B, \pi_G]$ whatever parents do $\blacktriangleright T_y(r) \in \mathcal{R}$
2. **Continuous:** u smooth, F_η has a density $\blacktriangleright$ offers $q_\theta(r)$ continuous, pool never empty
3. **Exists:** $T_y(r) - r \geq 0$ at the bottom of $\mathcal{R}$, ≤ 0 at the top $\blacktriangleright$ a fixed point in between
4. **Unique:** $T'_y = -\frac{(1 + \rho + g) \Delta\pi}{\mu_{co}^2} \frac{\partial \Lambda_1}{\partial r}$; offers respond to r by at most $\sup f_\eta \cdot \beta \pi_\theta u_c L^*$
5. $\blacktriangleright |T'_y| \leq C \sup f_\eta < 1$ if taste shocks are dispersed $\blacktriangleright$ contraction, iteration converges

[▶ next: cutoffs](#)

[◀ back](#)

Backup: proof, the two cutoffs

$\Delta V(y, \theta) = \pi_\theta G(y) + (1 - \pi_\theta) \Delta \underline{u}$ G : gain if she repays; $\Delta \underline{u} > 0$: she keeps the home

1. **G falls with income:** the solo house grows with y , the cosigned one is capped $\blacktriangleright dG/dy < 0$
2. **Single crossing:** $\frac{\partial^2 \Delta V}{\partial y \partial \pi_\theta} = \frac{dG}{dy} - \frac{d\Delta \underline{u}}{dy} < 0$ $\blacktriangleright$ the gain falls faster for safe children
3. **Order:** at the safe child's cutoff, $\Delta V(B) - \Delta V(G) = \Delta \pi (\Delta \underline{u} - G) > 0$ $\blacktriangleright \tilde{y}_B > \tilde{y}_G$
4. **Exist:** $\Delta V > 0$ at low income, < 0 at high income, continuous $\blacktriangleright$ a root
5. **Unique:** ΔV decreasing through its zero $\blacktriangleright$ one root per type

Backup: proof, the three regions are an equilibrium

1. **Low income:** $\Delta V(B) > \Delta V(G) > 0$, both types cosign if offered ► $\mu_{co} = \bar{\pi}_1$
2. **Middle:** safe children have left ► $\mu_{co} = \pi_B$, rate r^{sep} ; risky stay until $\Delta V(B; r^{sep}) = 0$
3. **High income:** nobody cosigns
4. **Deviations:** a safe child cosigning above her cutoff pays $r^{sep} > r^{pool}$ she declined
5. **Deviations:** solo at low income gives up $\Delta V > 0$; an unused offer gains the parent nothing
6. **Unique:** rates by break-even, cutoffs by the two crossings, beliefs by Bayes' rule

Backup: sorting, formally (the three regions)

Two cutoffs in the child's income, where each type stops gaining from the cosigned contract:

$$\Delta V(y_G^{pool}, G; r_{co}^{pool}) = 0, \quad \Delta V(y_B^{sep}, B; r_{co}^{sep}) = 0, \quad y_F < y_G^{pool} < y_B^{sep}$$

Cosigned rate and belief in each region:

$$1 + r_{co}(y) = \begin{cases} (1 + \rho + g)/\bar{\pi}_1(y) & y < y_G^{pool} : \text{both types cosign; screen } q_G > q_B \\ (1 + \rho + g)/\pi_B & y_G^{pool} \leq y < y_B^{sep} : \text{only risky children cosign} \\ \text{no cosigned contract} & y \geq y_B^{sep} \end{cases}$$

Needs: u concave with $u_{ch} > 0$, dispersed taste shocks, the key assumption, and risky children still cosigning at y_G^{pool} (else two regions)

Backup: what the signature reveals, by parent income

Belief of a lender who could read the parent's income y_p , with offers $q_\theta = F_\eta(a_\theta)$:

$$\Lambda_1(y, y_p) = \frac{\lambda q_G}{\lambda q_G + (1 - \lambda) q_B}, \quad a_G - a_B = \Delta\pi [\Delta u_p(y_p) - \beta (\Delta \underline{u} - G)]$$

- Parent covers part ($\bar{p} < 1$): a richer parent has more at stake ► screen **stronger**
- Parent covers all ($\bar{p} = 1$): paying hurts a richer parent less ► Λ_1 **falls** in y_p
- The rate ignores y_p ► families with poorer parents overpay, richer ones underpay
- Very rich parent, $\Delta u_p \rightarrow 0$: $q_G < q_B$, the guarantee is **bad news** (excluded)

Estimated model: cosigned default does not rise with parent income (untargeted moment)

Backup: the two admission tests, formally

A loan L on a house worth P is made under contract k only if **both** tests hold:

$$\underbrace{(1 + r_k) L \leq \overline{\text{dti}} Y_k, \quad Y_{so} = y, \quad Y_{co} = y + y_p}_{\text{income test}}$$

$$\underbrace{(1 - \overline{\text{ltv}}^k) P \leq a}_{\text{deposit, from the child's own savings } a}$$

- The parent enters the **income test** only, through Y_{co} (Fact 2)
- **Deposit:** no y_p , no rate; cap $\overline{\text{ltv}}$ of 95% (90% manual) with a cosigner, 97% solo
- Pricing cannot undo rationing: no rate admits a buyer the deposit excludes
- Guarantee counted toward the deposit: admits exactly those buyers ► the reform

Backup: the pricing rule, formally

The rate reads y and the contract k , not the parent: neither y_p nor her cover $\bar{p}(y_p)$.

Free entry: it breaks even on what the **child** is expected to pay (ρ funding cost, g fee)

$$\mu_k(y) (1 + r_k(y)) = 1 + \rho + g \quad \implies \quad 1 + r_k(y) = \frac{1 + \rho + g}{\mu_k(y)}$$

Belief: the child's own repayment probability in the pool taking contract k (Bayes' rule)

$$\mu_k(y) = \pi_B + \Delta\pi \Pr(G \mid y, k), \quad \Pr(G \mid y, co) = \frac{\lambda \bar{q}_G}{\underbrace{\lambda \bar{q}_G + (1 - \lambda) \bar{q}_B}_{\text{where both types cosign}}}$$

- Need and screen move the offer rates $\bar{q}_G, \bar{q}_B$: they are **in the rate**
- Lender collects $[\mu_{co} + (1 - \mu_{co}) \bar{p}(y_p)](1 + r_{co})$: the cover is **not in the rate**
- Not competed away: agency fee ignores the parent; off-agency loans lose liquidity

Backup: need and screen, formally

Child's gain: G if she repays, $\Delta \underline{u} > 0$ if she misses (she keeps the home)

$$\Delta V(y, \theta) = \pi_{\theta} G(y) + (1 - \pi_{\theta}) \Delta \underline{u}, \quad \pi_{\theta} = \text{repayment probability}, \quad \pi_G > \pi_B$$

Parent offers if altruism net of expected liability beats a taste shock $\eta_p \sim F_{\eta}$

$$q_{\theta}(y, y_p) = F_{\eta}(\beta \Delta V(y, \theta) - (1 - \pi_{\theta}) \Delta u_p), \quad \Delta u_p = u(y_p) - u(y_p - \text{payment she covers})$$

- **Need:** $\Delta V(y, B) - \Delta V(y, G) = \Delta \pi (\Delta \underline{u} - G) > 0$ where insurance outweighs G
- **Key assumption:** $\Delta u_p > \beta (\Delta \underline{u} - G) \triangleright q_G > q_B$, the **screen**
to the parent, a riskier child adds more liability than altruistic gain
- Fails for very rich parents, $\Delta u_p \rightarrow 0$: there the guarantee is bad news

Backup: why the net of need and screen is ambiguous

- **Low income:** both types cosign; only the screen sorts ► cosigned pool **safer**
- **Middle:** safe children leave first; only need sorts ► cosigned pool **riskier**
- **High income:** nobody cosigns

$$\bar{\mu}_{so} - \bar{\mu}_{co} = \underbrace{\int_{y_G}^{\infty} \bar{\pi} dF_{so} + \int_{y_G}^{y_B} (\bar{\pi}_2 - \bar{\pi}) dF_{so} + \int_{y_G}^{y_B} (\bar{\pi} - \pi_B) dF_{co}}_{\text{need} > 0: \text{ safe children leave; the solo pool gets cleaner, the cosigned pool risky}}$$

need > 0: safe children leave; the solo pool gets cleaner, the cosigned pool risky

$$- \underbrace{\left[\int_0^{y_G} (\bar{\pi}_1 - \bar{\pi}) dF_{co} + \int_0^{y_G} (\bar{\pi} - \bar{\pi}_0) dF_{so} \right]}_{\text{screen, subtracted: an offer is good news, no offer is bad news}}$$

screen, subtracted: an offer is good news, no offer is bad news

$$+ \underbrace{\int_0^{y_G} \bar{\pi} (dF_{so} - dF_{co}) - \int_{y_G}^{y_B} \bar{\pi} dF_{co}}_{\text{weights} \leq 0: \text{ the two pools are different sizes}}$$

weights ≤ 0: the two pools are different sizes

y_G, y_B : exits of safe, risky; $\bar{\pi}$: all buyers; $\bar{\pi}_1, \bar{\pi}_0$: with, without an offer; $\bar{\pi}_2$: solo, middle; $dF_k = \omega_k f_k dy$

► Sign from the data: model +0.28 pp; 30+ late, same cell, -0.24 (n.s.)

► rate schedule

◀ back

Backup: prediction 2, formally

A fee increase dg raises the posted rate by $dg/\mu_k(y)$, and the debt-to-income ratio by

$$dDTI_k = \sigma_k dg, \quad \sigma_k \equiv \frac{L}{Y_k} \cdot \frac{1}{\mu_k(y)}, \quad Y_{so} = y, \quad Y_{co} = y + y_p$$

Rejection probability at ratio d is $H_k(d)$; the parent is a compensating factor,

$$H_{co}(d) = H(d - \tau \gamma(y_p)), \quad H_{so} = H$$

$$\beta_{rej}^k \equiv \frac{\partial \Pr(\text{reject} \mid k)}{\partial g} = \mathbb{E}^k [H'_k(DTI_k) \sigma_k]$$

- **Rates:** both pools pay, $\beta_{rate}^k = 1/\mu_k$; predicted difference under 1 bp
- **Rejections:** $\beta_{rej}^{co} < \beta_{rej}^{so}$ once $\tau \geq \tau^*$, even if the cosigned rate moves more
- Fact 2: at the same ratio cosigned files are rejected less, 7.1 vs. 9.8% at 45–49

◀ model

◀ predictions

Backup: prediction 1, derivation (rates)

1. Break-even rate on contract k : $1 + r_k(y) = \frac{1 + \rho + g}{\mu_k(y)}$ (g : agency fee)
2. A fee change moves both rates: $\beta_{rate}^k(y) \equiv \frac{\partial r_k(y)}{\partial g} = \frac{1}{\mu_k(y)}$
3. The pools differ only through pool risk: $\beta_{rate}^{co} - \beta_{rate}^{so} = \frac{\mu_{so}(y) - \mu_{co}(y)}{\mu_{co}(y) \mu_{so}(y)}$
4. Pools about alike (Fact 4) ► the difference is a small share of dg : under 1 bp for the 2022 fee change
5. The parent's payments $\bar{p}$ enter nowhere ► the rate side cannot show the cure

Backup: prediction 2, derivation (admission)

1. $DTI_k = (1 + r_k) L / Y_k$, $Y_{so} = y$, $Y_{co} = y + y_p$; fee: $dDTI_k = \sigma_k dg$, $\sigma_k = L / (Y_k \mu_k)$
2. Rejection probability $H_k(DTI_k)$ $\blacktriangleright \beta_{rej}^k \equiv \partial \mathbb{E}^k[H_k] / \partial g = \mathbb{E}^k[H'_k(DTI_k) \sigma_k]$
3. Compensating factor: $H_{so} = H$, $H_{co}(d) = H(d - \tau \gamma(y_p))$, H convex
4. $\Phi(\tau) \equiv \mathbb{E}^{co}[H'(DTI_{co} - \tau \gamma) \sigma_{co}]$ falls in τ ; at $\tau = 0$: cosigned rejected **more**
5. $\blacktriangleright$ one threshold: $\beta_{rej}^{co} < \beta_{rej}^{so} \iff \tau > \tau^*$ (Fact 2: flatter cosigned hazard)

[statement](#)

[back](#)

Backup: the unpriced guarantee, formally

Posted rate on the child's own repayment probability $\mu_k(y)$, fair rate with parental coverage $\bar{p}(y_p)$:

$$1 + r_k(y) = \frac{1 + \rho + g}{\mu_k(y)}, \quad 1 + \tilde{r}_{co}(y, y_p) = \frac{1 + \rho + g}{\mu_{co}(y) + (1 - \mu_{co}(y))\bar{p}(y_p)}$$

Fair cosigned rate below the solo rate if and only if

$$\bar{p}(y_p) > \bar{p}^*(y) \equiv \frac{\mu_{so}(y) - \mu_{co}(y)}{1 - \mu_{co}(y)}$$

- Wedge $r_{co} - \tilde{r}_{co}$ rises with coverage and with pool risk
- No rate, however low, admits a borrower the deposit excludes

Backup: robustness, if the lender ignores the pool

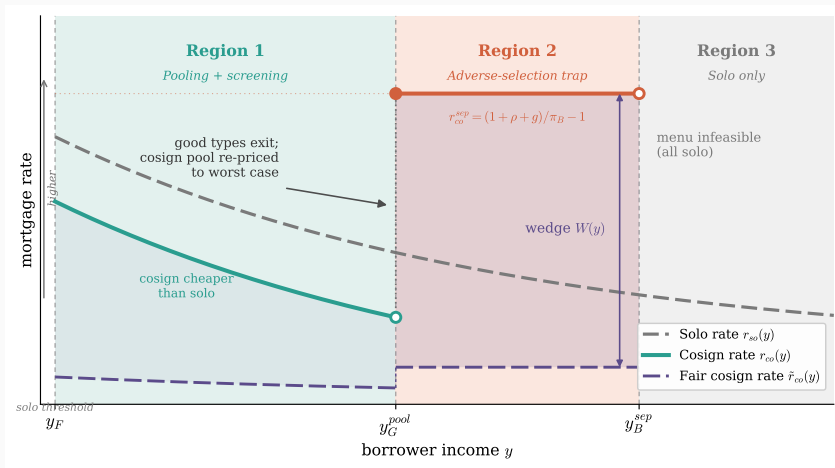
Result holds: the fair cosigned rate stays **below** the rate charged, even if the rate ignores who cosigns (agencies' grid: score and loan-to-value only)

The rate considers the pool with weight α , and all buyers' repayment $\bar{\pi}$ otherwise:

$$1 + r_k = \frac{1 + \rho + g}{\alpha \mu_k + (1 - \alpha) \bar{\pi}} \quad \text{holds if and only if} \quad \bar{p} > (1 - \alpha) \frac{\bar{\pi} - \mu_{co}}{1 - \mu_{co}}$$

	$\alpha = 1$: the paper	$\alpha = 0$: grid only
Parent must cover more than	0	at most $\bar{p}^* \approx 0$ (Fact 4)
Pool feeds back into the rate	yes	no
Separating rate	yes	no

Backup: the rate schedule across borrower income



Backup: the 2022 fee change, institutional details

- **Who:** the FHFA, regulator of Fannie Mae and Freddie Mac
- **When:** announced 5 January 2022 (the cutoff used), in force 1 April 2022
- **What:** higher upfront fee on high-balance loans, by loan-to-value ratio **alone**
- **Which fee:** the risk-based part of the guarantee fee, paid once at sale
- **The grid:** credit score and loan-to-value; no cosigner, no income
- **Size:** 25 to 75 bp of the loan ▶ at most 0.19 pp of note rate
- **Which loans:** \$647,200 up to the county ceiling (at most \$970,800); jumbo untouched
- **Exposure:** cosigned loans are sold to the agencies more often than solo loans

[▶ fee schedule](#)[▶ sample and design](#)[◀ back](#)

Backup: sample and design details

Each pool k separately; year effects and slopes, controls, lender effects:

$$Y_{ilt}^k = \alpha_{year}^k + \beta_1^k D_{it} + \beta_3^k (D_{it} \times Post_t) + f_{year}^k(\tilde{X}_{it}) D_{it} + Z_{it} \gamma^k + \delta_l^k + \epsilon_{ilt}^k$$

- β_1 : jump at the limit before the reform; β_3 : **how much the jump changed**
- Loan size relative to that year's limit; seven yearly limits stacked, 2018–24
- After = 2022 onward, from the January 2022 announcement
- HMDA loan size in \$5,000 brackets ► cutoff at the first bracket above the limit
- Income above 120% of area median: below it, first-time buyers' fee could be waived
- Bunching below the limit: after vs. before cancels it if it did not change

Backup: what identifies the difference-in-discontinuities

Four cells, below or above the limit (s) by before or after (t), for each pool k :

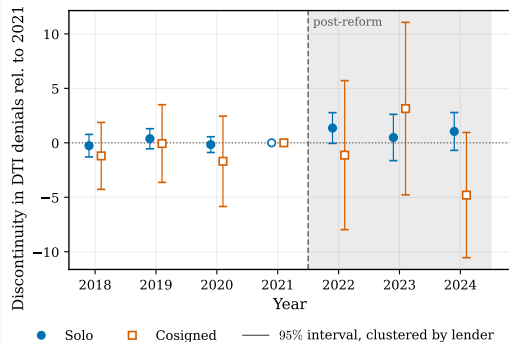
$$1 + R_{s,t}^k = \frac{1 + \rho_t + g_{s,t}}{\mu_{s,t}^k}, \quad g_{+,Post} = g_0 + \underbrace{\Delta g^{HB}}_{\text{old fee}} + \underbrace{\Delta g^{reform}}_{2022}, \quad \rho_{Post} = \rho + \underbrace{\Delta \rho}_{\text{rate cycle}}$$

1. **Constant composition**, $\mu_{s,Post}^k = \mu_{s,Pre}^k$: removes the old jump and the sorting
 2. **Continuity in differences**: the rate cycle moves both sides alike, $\mu_+^k \approx \mu_-^k$
 3. **No switching**: the reform moves nobody into cosigning (separate pools)
 4. **Additivity**: new jump = old jump + reform effect; the two do not interact
- $\beta_3^k = \Delta g^{reform} / \mu^k$. Solo vs. cosigned: 2 weakens to similar sorting in both pools

Backup

Checks: no pre-trends; stable, similar bunching; placebo cutoffs; no contract switching

Backup: condition 1, composition at the limit constant over time



- **No pre-trends:** the yearly jump in denials is flat 2018–21
joint test $p = 0.76$ solo, 0.90 difference
- **Borrowers:** no **differential** jump in income, age, gender
cosigned – solo, $p > 0.10$; solo income alone -0.053 (5%)
- **Bunching:** density jump unchanged
 -2.07 ($p = 0.54$) solo, -1.95 ($p = 0.58$) cosigned

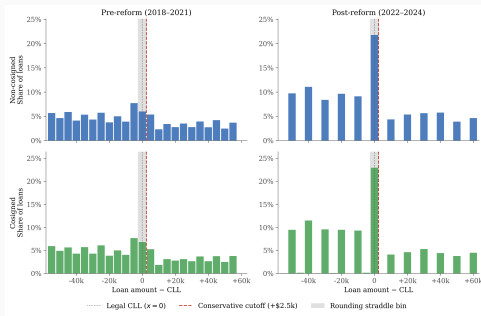
► loan-size density

► next: condition 2

► identification

◀ back

Backup: loan-size density around the limit



The shaded bar straddles the limit and distorts the LEVEL of the jump; common to both periods, it cancels in the difference

► the test

► condition 1

◄ back

Backup: the difference-in-McCrary test

A plain McCrary rejects here by construction: bunching is real and the \$5,000 bracket distorts its size. Test the **change** instead, bin by bin:

$$s_{b,t} = \alpha_{year} + \beta_1 above_b + \beta_3 (above_b \times Post_t) + f(x_b) + \varepsilon_{b,t}$$

$s_{b,t}$ = share of the year's loans in \$1,000 bin b ; $above_b$ = bin wholly above the limit (+\$2,500); f = own slope each side, each period; HC3 errors

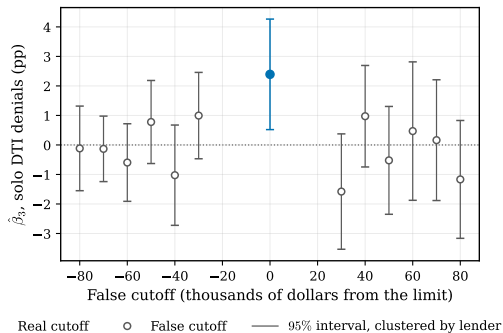
	$\hat{\beta}_3$ (above $\times$ post)	SE	p
Solo	-2.07	3.40	0.542
Cosigned	-1.95	3.50	0.576

► The heap cancels in the difference; **intensity did not change**

► the density

◄ back

Backup: condition 2, continuity at the cutoff in differences



- **False cutoffs:** no change in the jump at \$30–80k from the limit; the real cutoff ranks first of thirteen ($p = 0.077$)
- Farther out, at \$70–160k, false jumps do appear ► the claim is local to the limit
- **False reform date:** no difference between the pools, -0.01 ($p = 0.93$) and $+0.03$ ($p = 0.79$)

Backup: condition 3, no switching into cosigning

- **Why it matters:** the two pools are estimated separately, so the reform must not change who is in them
- **Why it could fail:** a guarantee rescues a file at the income test, so a solo applicant near the cap might bring in a parent
- **Test:** change in the probability that a loan at the limit is cosigned: -0.000
- **Why it is plausible:** the fee is at most 0.19 pp of note rate, a small reason to ask a parent to sign
- An unchanged share rules out net switching; borrower characteristics at the limit are also stable in each pool (condition 1)

Backup: condition 4, additivity

New jump = old jump **plus** the reform's effect, with no interaction:

$$1 + R_{+,Post}^k = \frac{1 + \rho + \Delta\rho + g_0 + \Delta g^{HB} + \Delta g^{reform}}{\mu^k}$$

- Fees **add** in the pricing rule: a flat amount on top of the old high-balance fee
- Sharp design: high-balance status switches exactly at the limit, before and after
- Would fail if the reform changed how lenders treat the **old** fee, not just its size
- Argued from the fee schedule, not tested (Picchetti et al. 2024)

Backup: why not a simple regression?

A simple comparison would pick up

Loans above the limit differ: size, income, lender

High-balance loans paid a fee before 2022 as well

Borrowers bunch below the limit to avoid the fee

Rates moved 300 bp; the limit resets every year

Cosigned and solo borrowers differ

What the design does about it

Compare loans **just** above and below the limit

Difference the jump, **after vs. before**

The same differencing cancels unchanged sorting

Year effects and slopes; seven limits stacked at zero

Each pool estimated **separately**, then compared

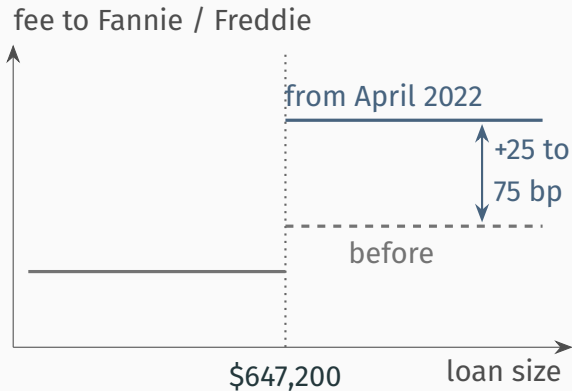
► Not hypothetical: an earlier pooled specification found a rate gap that a placebo outcome also showed; year effects flipped its sign

► sample and design

► validity

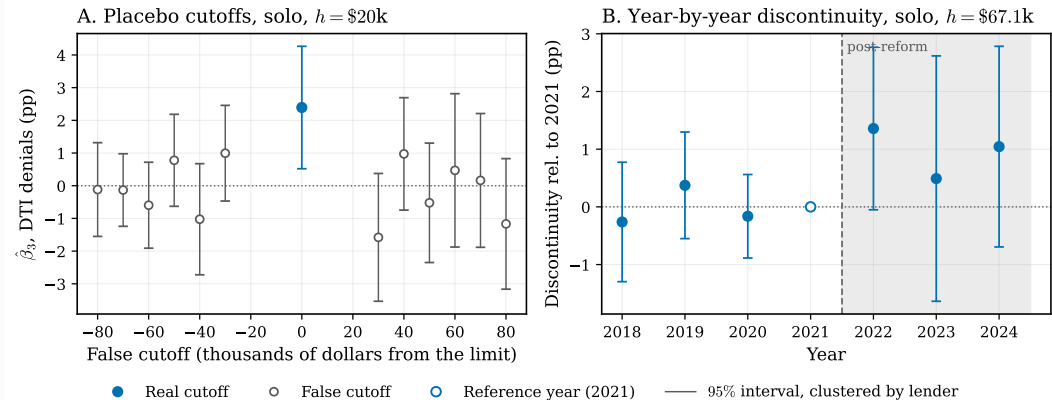
◀ back

Backup: the 2022 fee change



Upfront fee on high-balance loans sold to Fannie Mae or Freddie Mac, tiered by loan-to-value ratio alone

Backup: natural experiment, validity



◀ back

Backup: natural experiment, more numbers

- Solo, optimal window (\$67.1k): +1.11 pp, $p = 0.018$; all windows +0.96 to +1.95
- Nearer the limit (\$20k): +2.39 pp
- Bias-aware interval for the discrete loan-size variable: [+0.37, +2.75]
- Cosigned: -0.11 pp, 95% interval [-5.15, +4.94]
- Rate, cosigned – solo: -0.07 pp, [-0.52, +0.39]; predicted difference under 1 bp
- Sample: income above 120% of area median (fee waiver below), jumbo loans excluded

Backup: income-test denials, full results

h (\$000)	Income above area median (primary)			Full sample		
	Solo $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	$\hat{\Delta}$	Solo $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	$\hat{\Delta}$
30	+1.73** (0.69)	+0.72 (3.84)	-1.02	+1.12 (0.77)	+1.03 (5.16)	-0.09
40	+1.95*** (0.59)	-0.71 (3.31)	-2.66	+0.91 (0.58)	-0.77 (3.69)	-1.68
50	+1.72*** (0.58)	-1.35 (2.84)	-3.07	+1.27** (0.52)	-0.95 (3.31)	-2.22
60	+1.30** (0.54)	-1.62 (2.70)	-2.92	+1.24*** (0.47)	-2.61 (3.08)	-3.85
67.1[†]	+1.11** (0.47)	-0.11 (2.58)	-1.22	+0.84* (0.48)	-1.84 (3.08)	-2.68
70	+1.08** (0.47)	+0.67 (2.53)	-0.40	+0.79* (0.47)	-1.08 (3.03)	-1.87
76.6 [†]	+0.96** (0.43)	-0.14 (2.08)	-1.11	+0.75 (0.46)	-1.82 (2.33)	-2.57
80	+0.96** (0.42)	-0.09 (2.00)	-1.05	+0.70 (0.46)	-1.86 (2.27)	-2.55
90	+1.05*** (0.40)	-0.16 (1.98)	-1.21	+0.83* (0.42)	-1.48 (2.16)	-2.31

Outcome: application denied with debt-to-income cited as a reason (HMDA), percentage points. Year-fixed-effects difference-in-discontinuities; standard errors in parentheses, clustered by lender. $\hat{\Delta}$ = cosigned - solo. [†] MSE-optimal windows, solo then cosigned. Samples over the paper's windows (40-90): 35,490-92,643 solo and 3,839-10,341 cosigned (primary); 76,206-213,082 and 5,147-14,433 (full). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Backup: interest rates, full results

h (\$000)	Solo $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	$\hat{\Delta}$	p_{Δ}^{WCB}	N solo	N cosigned
18	+0.0162 (0.1192)	+0.0075 (0.2403)	−0.0087 (0.2271)	0.97	22,453	2,433
20	−0.0097 (0.1136)	−0.0661 (0.2389)	−0.0563 (0.2331)	0.82	23,985	2,586
20.8[†]	−0.0098 (0.1133)	−0.0772 (0.2388)	−0.0674 (0.2318)	0.78	24,809	2,676
22	−0.0071 (0.1133)	−0.0623 (0.2351)	−0.0552 (0.2301)	0.82	25,682	2,768
24	−0.0162 (0.1145)	−0.0625 (0.2332)	−0.0463 (0.2297)	0.84	26,882	2,880
26	−0.0073 (0.0888)	−0.2047 (0.1724)	−0.1974 (0.1813)	0.28	30,751	3,328
28	+0.0004 (0.0863)	−0.1630 (0.1650)	−0.1634 (0.1593)	0.31	32,037	3,476

Outcome: contract interest rate, percentage points. Same specification, on the primary sample (income above area median); covariates and lender fixed effects; standard errors in parentheses, clustered by lender. $\hat{\Delta}$ = cosigned − solo; p_{Δ}^{WCB} = wild-cluster-bootstrap p-value of the differential. [†] the solo pool's MSE-optimal window: −0.07 pp, 95% interval [−0.52, +0.39], against a predicted difference under 1 bp. No estimate is significant at 10%.

[► denials table](#)[◄ back](#)

Backup: event study, full results

Year	$h = \$20k$		$h = \$67.1k$ (slide)		$h = \$90k$	
	Solo	Cosigned	Solo	Cosigned	Solo	Cosigned
2018	-0.80 (1.14)	-7.29 (4.21)	-0.26 (0.53)	-1.20 (1.57)	+0.22 (0.42)	-3.53 (1.78)
2019	+0.22 (1.04)	-5.55 (8.12)	+0.37 (0.47)	-0.07 (1.82)	-0.06 (0.47)	-1.65 (1.82)
2020	-0.32 (0.78)	-10.19 (5.48)	-0.16 (0.37)	-1.70 (2.12)	-0.17 (0.30)	-1.45 (1.48)
2021	ref.	ref.	ref.	ref.	ref.	ref.
2022	+2.88 (1.63)	-6.95 (6.79)	+1.36 (0.72)	-1.13 (3.49)	+1.46 (0.58)	-1.18 (2.60)
2023	+1.21 (2.27)	+6.53 (13.69)	+0.49 (1.09)	+3.14 (4.04)	+0.05 (0.99)	-1.30 (3.26)
2024	+0.84 (1.89)	-11.53 (5.63)	+1.04 (0.89)	-4.80 (2.93)	+1.09 (0.85)	-3.12 (2.89)
Pre-reform Wald p	0.89	diff. 0.35	0.76	diff. 0.90	0.84	diff. 0.23

Yearly discontinuity in debt-to-income-cited denials at the conforming limit, relative to 2021, percentage points; standard errors in parentheses, clustered by lender. Last row: joint Wald test that the 2018–2020 coefficients are zero, for the solo pool and for the cosigned-minus-solo difference. Loans per year at $h = \$67.1k$: 5,069–14,851 solo, 561–1,785 cosigned.

Backup: false cutoffs, full results

False cutoff (\$000 from limit)	$h = \$20k$ (slide)			$h = \$24k$		
	$\hat{\beta}_3$	t	N	$\hat{\beta}_3$	t	N
-80	-0.12 (0.73)	-0.16	19,926	+0.11 (0.67)	+0.16	22,888
-70	-0.13 (0.57)	-0.23	20,328	-0.23 (0.54)	-0.43	23,344
-60	-0.60 (0.67)	-0.89	20,742	-0.49 (0.69)	-0.72	23,921
-50	+0.78 (0.72)	+1.08	20,968	+0.70 (0.72)	+0.96	24,288
-40	-1.02 (0.87)	-1.18	21,392	-0.96 (0.84)	-1.14	24,768
-30	+0.99 (0.75)	+1.33	21,913	+1.01 (0.71)	+1.41	25,498
0 (real)	+2.39 (0.96)	+2.50	25,296	+2.38 (0.95)	+2.51	28,354
+30	-1.58 (1.00)	-1.58	18,395	-1.68 (0.93)	-1.80	21,080
+40	+0.97 (0.88)	+1.11	19,012	+0.88 (0.85)	+1.03	21,919
+50	-0.52 (0.93)	-0.56	18,799	-0.17 (0.92)	-0.19	21,535
+60	+0.47 (1.20)	+0.39	17,725	+0.40 (1.18)	+0.34	20,505
+70	+0.16 (1.04)	+0.16	16,536	+0.00 (1.06)	+0.00	19,266
+80	-1.17 (1.02)	-1.15	15,594	-1.09 (1.00)	-1.09	18,291

Solo pool, debt-to-income-cited denials, pp; results-slide specification with the limit moved as stated; standard errors (in parentheses) clustered by lender. The real cutoff has the largest $|t|$ of thirteen at both windows: exceedance $p = 1/13 = 0.077$. Farther out (\$70–160k) false jumps do appear: the claim is local.

Backup: model details

- Ages 18–88; persistent income shocks, independent across generations
- One purchase decision, at 36: “ownership” = owning by the mid-thirties
- 30-year mortgage; each year pay or default, with taste shocks
- Default costs a utility penalty, lower for risky types; lower still when a parent is called on
- Parent pays the missed payment as long as her income allows
- Lender loses 25–40% of the balance on a foreclosure
- Housing supply elasticity 2.2; rental and owner markets clear

Backup: the child's problem

Away from the two decision ages, consumption and savings under survival risk:

$$V_j(\mathbf{s}_j) = \max_{a' \geq 0} \left\{ u(c_j, h_j, \mathbb{I}_{own}) + \beta s_j \mathbb{E}[V_{j+1}(\mathbf{s}_{j+1})] + \beta(1 - s_j) \Psi(b_{j+1}) \right\}$$

$$c_j + a' + \underbrace{\Gamma(h_j, \kappa_j, \mathbf{s}_j)}_{\text{rent } p_r h, \text{ payment } m, \text{ upkeep } \delta_h p_{own} h} = w e(z_j, j) + (1 + r)a_j + \underbrace{Tr_j(z_{p,j})}_{\text{inheritance}}$$

- $\mathbf{s} = \{j, a, z, z_p, \omega, \kappa, \zeta, h\}$: age, assets, both incomes, tenure, help type, exclusion, house
- u Cobb–Douglas over c and h , CRRA, ownership premium $\nu > 1$; Ψ warm-glow bequest
- The decision ages are on the equation slides: take-up at 36, helping at 61

Backup: the parent's problem

At the helping age the parent picks none, cosign, cash or both under a taste shock:

$$V_{43}(\mathbf{s}, \theta) = \sigma_{help} \ln \sum_{\kappa=1}^4 \exp\left(\frac{V_{43}^p(\mathbf{s}, \theta, \kappa) + \lambda_{\kappa}(z_p) \Delta V_c(\theta, z_p, \kappa) + \eta_{\kappa}}{\sigma_{help}}\right)$$

- V^p : own value, carrying the cash transfer **and** the expected liability
- Expects to pay $(1 - \pi_{\theta})m$, **more for a Bad child** ▶ $q_G > q_B$ is a **result**, not an assumption
- $\lambda_{\kappa}(z_p) = \bar{\lambda}_{\kappa}(e(z_p, 43)/e(\bar{z}_p, 43))^{\gamma}$: altruism scaled by own income
- ΔV_c : the child's gain, over the parent's belief about its income

Backup: the lender's problem

No maximisation: free entry prices each pool to break even on what the rule can see.

$$\underbrace{\frac{m}{y(z_c)} \leq \bar{\omega}_{solo} = 0.40}_{\text{solo}} \quad \underbrace{\frac{m}{y(z_c)+y(z_p)} \leq \bar{\omega}_{joint} = 0.42 \text{ and } \frac{m}{y(z_c)} \leq \bar{\omega}_{own} = 1.0}_{\text{cosigned}}$$

- The income test moves with a signature; the **deposit does not**, $(1 - \lambda_m) p_{own} h$
- Rate: base $r_{base}(z_c)$ plus $\Delta r(h, z_c)$, break-even on each pool hazard
- Sees h, z_c and the contract ► never the type θ , nor what the family knows
- $\bar{\omega}_{own} = 1.0$: the only solo test on a cosigned file, it blocks certain default

► market clearing

► rate schedule

► pricing rule

◄ back

Backup: what the parent actually covers

On a **cosigned** default the child keeps home and mortgage; the parent pays what income allows:

$$p_{cover}(h, z_p, z_c) = \min \left\{ 1, \frac{y(z_p)}{m(h, z_c)} \right\}$$

- m the annual payment, $y(z_p)$ parental income at the helping age
- A parent who falls short gives up **all** of it ▶ costlier for **low-income parents**
- One year of debt service is the smallest obligation implied ▶ a **normalisation**, not an estimate
- If the parent cannot cover: foreclosure, the child rents permanently (anticipated at take-up)

Backup: what the parent knows, and the lender does not

At the helping age the parent sees its own state and the child's **type** θ , but **not** the child's income:

$$W_{\theta}(z_c) \propto \Phi(z_c) \lambda(z_c)^{\mathbb{I}\{\theta=G\}} (1 - \lambda(z_c))^{\mathbb{I}\{\theta=B\}}$$

- Φ the income distribution; $\lambda(z_c)$ the prior that the child is Good, **rising in income**
- The parent's belief tilts toward the incomes at which its child's type is common
- ΔV_c is taken over this belief, and over the child's initial assets
- The lender holds the same prior $\lambda(z_c)$ but sees neither θ nor the family's information ► the **screen works where pricing cannot**

Backup: the model appendix, slide by slide

The child

- the problem
- preferences
- income
- default penalties
- model details

The parent

- the problem
- what she knows
- the type prior
- what she covers

The lender and the market

- the problem
- rate schedule
- pricing rule
- market clearing
- construction
- equilibrium
- convergence
- from the static model

Estimation

- identification
- estimates
- all parameters
- fit at a glance

Backup: preferences and bequests

Cobb–Douglas over consumption and housing services, CRRA curvature:

$$u(c, h, \mathbb{I}_{own}) = \frac{[c^{1-\alpha} (\nu(\mathbb{I}_{own})h)^\alpha]^{1-\sigma} - 1}{1-\sigma}, \quad \nu(\mathbb{I}_{own}) > 1 \text{ if owner, } = 1 \text{ if renter}$$

Warm glow over the estate, De Nardi 04:

$$\Psi(b) = \phi_1 \frac{(b + \phi_2)^{1-\sigma} - 1}{1-\sigma}, \quad b = a' + (1 - \delta_h) p_{own} h$$

- α the housing expenditure share; $\nu > 1$ is the **ownership premium**
- ϕ_1 the strength of the motive, ϕ_2 its curvature ► bequests are a **luxury good**
- The estate is financial assets plus the depreciated house

Backup: the construction sector

Competitive builders, constant returns in land and structures:

$$H_s(p) = C p^\varepsilon, \quad \varepsilon = \frac{\alpha_c}{1 - \alpha_c} = 2.2$$

- **Baseline:** p_{own} is calibrated to the ownership rate ▶ the same as **perfectly elastic** supply
- **GE leg:** both markets clear, $H_d^{own} = C_{own} p_{own}^\varepsilon$ and $H_d^{rent} = C_{rent} p_{rent}^\varepsilon$
- ε from the land share in first-time buyers' purchases (FHFA)
- The reform's ownership gain shrinks as ε falls ▶ the elasticity is the GE lever

▶ gain by elasticity

▶ market clearing

▶ the lender

◀ back

Backup: market clearing

Credit — cell by cell in (h, z_c) , on the composition at origination:

$$\Delta r(h, z_c) = \frac{R(z_c)}{\mu_{co}(h, z_c)} - \frac{R(z_c)}{\mu_{so}(h, z_c)}, \quad \mu_k = \tilde{\pi}_B + (\tilde{\pi}_G - \tilde{\pi}_B) q_G^k$$

q_G^k is an **equilibrium** object for **both** pools, never the prior $\lambda(z_c)$. Thin cells keep the previous premium; the matrix updates **damped**, at $\frac{1}{2}$.

Housing — (p_{own}, p_{rent}) clears both markets against iso-elastic supply $H_s(p) = C p^\varepsilon$:

$$H_d^{own}(p_{own}, p_{rent}) = C_{own} p_{own}^\varepsilon, \quad H_d^{rent}(p_{own}, p_{rent}) = C_{rent} p_{rent}^\varepsilon$$

Solved **jointly in logs**: $x = (\log p_{own}, \log p_{rent})$,

$F(x) = (\log H_d^{own} - \log H_s^{own}, \log H_d^{rent} - \log H_s^{rent})$; safeguarded Broyden, $\|F\|_\infty < 10^{-3}$ in 7–9 solves. Baseline = prices fixed, the $\varepsilon \rightarrow \infty$ case.

Inheritance — estates pooled by (z_p, z_c) , paid to children at $j = 48$:

$$Tr_{inherit}(z_p, z_c) = \frac{1}{\Omega} \sum_{j=G}^J \int (1 - s_j) b_j(\mathbf{s}_j) \mathbb{1}\{z_{c,j} = z_c\} d\mathcal{M}_j(\mathbf{s}_j | z_p)$$

Backup: consistency conditions on parental help

Help type — beliefs at $j = 18$ match what parents do at $j = 43$, for $\kappa \in \{\text{Cosign, Cash, Both}\}$:

$$\pi_{\kappa}(z_p, z_c) = \frac{\int \mathbb{1}\{\text{parent chooses } \kappa\} d\mathcal{M}_{43}(\mathbf{s}_{43} \mid z_p, z_c)}{\int d\mathcal{M}_{43}(\mathbf{s}_{43} \mid z_p, z_c)}$$

Each type also carries a **house-size** distribution π_{κ}^h , for cosigning resolved **by type** ($\pi_{\text{cos},\theta}^h$): the two types buy differently.

Cash — a fraction of the *parent's* income at $j = 43$, scaled by house size (not of the house value):

$$\bar{\psi}_{\text{cash}}(z_p, h) = \psi_{\text{cash}} \cdot w \cdot e(z_p, 43) \cdot h$$

Cosigning liability — carried in the parent's budget in expectation, child and house integrated out:

$$L(z_p) = \mathbb{E}_{z_c, h} \left[\varphi \bar{d}(z_p, z_c, h) p_{\text{cover}}(h, z_p, z_c) m(h, z_p, z_c) \right]$$

Backup: the help transfers add up

Cash — paid by parents at $j = 43$, received by children at $j = 18$:

$$\underbrace{\int \bar{\psi}_{cash} \mathbb{1}\{\kappa \in \{\text{Cash, Both}\}\} d\mathcal{M}_{43}}_{\text{given}} = \underbrace{\int \psi_{cash} w e(z_p, 43) h \mathbb{1}\{\text{buys with cash}\} d\mathcal{M}_{18}}_{\text{received}}$$

Cosigning — the counterparty is the **lender**, not the child:

$$\underbrace{\sum_{j=44}^{73} \int L(z_p) \mathbb{1}\{\kappa \in \{\text{Cosign, Both}\}\} d\mathcal{M}_j}_{\text{paid by parents}} = \underbrace{\sum_{j=19}^{18+M} \int \varphi p_{cover} m \mathbb{1}\{\text{cosigned miss}\} d\mathcal{M}_j}_{\text{collected on the mortgage}}$$

Cash is bilateral, so it clears inside the two budget constraints — no pooling, and no $\frac{1}{\Omega}$.
The cosigned flow is the -52 to -84 bp the rate never prices.

Backup: what the stationary equilibrium requires

- Households optimise given prices, the rate schedule and their beliefs
- **Beliefs are right:** $\pi_{\kappa}(z_p, z_c)$ matches how often parents actually choose κ
- ... and the **house-size distributions** behind them, **by type** for cosigning
- The cash cost and the **expected liability** match the policies they induce
- Premiums $\{\Delta r(h, z_c)\}$ match the pool compositions they cause
- Inheritance flow = the windfall children get; in the GE leg both housing markets clear

Every block updates **damped**: the pool-composition block has an unstable best response, so an undamped update **cycles**

Backup: the type prior $\lambda(z_c)$

Logistic in log income relative to the median at the purchase age:

$$\lambda(z_c) = \Pr(\theta = G \mid z_c) = \left[1 + \exp(-\lambda_\alpha \log[e(z_c, 18)/\tilde{e}]) \right]^{-1}$$

- $\lambda' > 0$, and $\lambda(\tilde{e}) = \frac{1}{2}$: the median earner is a coin flip
- Informative only through **relative** position in the income distribution
- $\lambda_\alpha = \sqrt{1 - \rho_z^2}/\sigma_\varepsilon \approx 2.55$: a one-sd income move shifts the log-odds by one
- Steep prior on a narrow grid = flat prior on a wide one ► this **fixes the units**
- With $\Delta\pi$, the **information primitives**, both set outside the estimation

► the parent's belief

► income process

◄ back

Backup: the income process

Labour income is $w e(z, j)$: a deterministic age profile times a persistent productivity state.

- Age profile P_j from the **PSID**: head labour income by three-year age group, 2003–2021
- Net of year means, non-retired heads (no retirement stage); model age $j = \text{age} - 18$
- z persistent, $(\rho_z, \sigma_\varepsilon)$, and **independent across generations**: the child draws from the stationary distribution
- The same $(\rho_z, \sigma_\varepsilon)$ pin the prior steepness λ_α

▸ the type prior

▸ the child

◀ back

Backup: the risk-based rate schedule

The base rate is a schedule on the persistent income state, applied before the pool premium.

- From the **NSMO rate-by-FICO gradient**, conventional first-time solo borrowers
- 4.75% at the bottom falling to 4.26% at the top (lowest states: no buyer mass)
- **Why z_c ?** The LLPA grid is scored on **score and LTV, not income**, and the model has no score state ► they are mapped **by rank**
- The two-track rule: **DTI gates admission, the score prices the loan**
- A DTI-priced schedule is a robustness leg (inert: ~95% of both pools in one bucket)

Backup: when the fixed point counts as solved

Max-norm change below tolerance on **every** block: default policy, premium, help probabilities, cash cost, pool composition. The map is **not a contraction**, so:

- **Inhabited-cell norm**: an empty cell can flip forever without unsettling the equilibrium (which norm certified a solve is recorded)
- **Consecutive** iterations: met once fits a **cycle** through the band as well as a fixed point
- States settle **before** the moments ► no certification while a reported moment still moves

A plateau rule aborts a solve whose inhabited norm stops improving

► equilibrium

► the lender

◄ back

Backup: which moment identifies which parameter

	M_1	M_2	M_3	M_4	M_5	M_6
Parameter	HO	Default	Cosign%	Cash%	Type ratio	Wedge
$\bar{\xi}$ penalty level	○	●				○
$\Delta\xi$ penalty spread		○			●	
div invocation penalty						●
p_{own} price/income	●	○				
η_{cosign}			●			
η_{cash}			○	●		

- primary, ○ secondary. Six parameters, six moments: **just-identified**

The standard errors in the moment table are **weighting tolerances** in the units of the target, not sampling errors

Backup: the default penalties

- π_θ : the **involuntary** floor; the lender prices it only through pool quality μ_k
- ξ_θ : the **stigma**, charged every time the child defaults, entering V^{def} directly and holding default down
- $\xi_{G,B} = \bar{\xi} \pm \Delta\xi/2$ with $\xi_G > \xi_B$ ► the **Good type suffers more**, which is what makes default type-informative
- $\kappa = \bar{\xi}/\text{div}$ per **invocation** on a cosigned default: the price of calling the parent
- $\bar{\xi}$, $\Delta\xi$ and div are all **estimated** (the penalty level, its spread, and the invocation penalty)

Backup: parameter estimates

Parameter	Symbol	Estimate
Default penalty, level	$\bar{\xi}$	4.8
Default penalty, type spread	$\Delta\xi$	4.99
Penalty divisor when the parent is called on	div	1.27
House price / income	p_{own}	12.26
Cost of cosigning	η_{cosign}	-1.10
Cost of a cash gift	η_{cash}	-0.53

► all parameters

◄ back

Backup: every parameter and where it comes from (1/3)

Parameter	Symbol	Value	Source
<i>From the literature, or fixed with literature motivation</i>			
Involuntary repayment gap	$\Delta\pi$	0.05	Gupta–Hansman 22
Altruism elasticity	γ	1.00	De Nardi 04; Lockwood 12 (conservative)
Discount factor	β	0.960	Cocco 05; Nakajima–Telyukova 20
Return on savings	r	0.030	Jordà et al. 19
Risk aversion	σ	1.50	Bacher 22 (low end of the range)
Housing expenditure share	θ	0.25	Davis–Ortalo-Magné 11
Housing depreciation	δ_h	0.015	Harding et al. 07
Income persistence	ρ	0.96	Heathcote et al. 10 (PSID)
Income innovation s.d.	σ_ϵ	0.110	Heathcote et al. 10
Age profile of income	P_j	PSID	PSID family files 2003–21

► institutional and internal

◄ back

Backup: every parameter and where it comes from (2/3)

Parameter	Symbol	Value	Source
<i>Institutional design</i>			
Loan-to-value limit, solo	λ_m	0.90	US conforming, 10% down; Benetton et al. 22
Loan-to-value limit, cosigned	λ_m^{co}	0.90	equal: NSMO shows no cosigned–solo gap
Recourse fraction	φ	1.00	co-obligor, jointly and severally liable
Mortgage term (years)	M	30	30-year fixed
Conforming-limit house tier	h^{CLL}	3	tier at which the limit binds
<i>Calibrated to an internal target</i>			
Bad-type repayment prob.	π_B	0.90	aggregate default level
Rent-to-price ratio	p_{rent}/p_{own}	0.090	quality-adjusted yield, Urban Institute 25
Owner utility premium	ν	1.100	the homeownership rate
Cash-transfer size	ψ_{cash}	0.100	PSID transfer distribution, Brandsås 25
Rental-grid cap	h_{rent}^{max}	0.40	makes $HO(z)$ monotone (cost: size ratio)
Type-income gradient scale	λ_α	2.545	normalisation, $\sqrt{1 - \rho^2}/\sigma_\epsilon$

► taste and numerical

◄ back

Backup: every parameter and where it comes from (3/3)

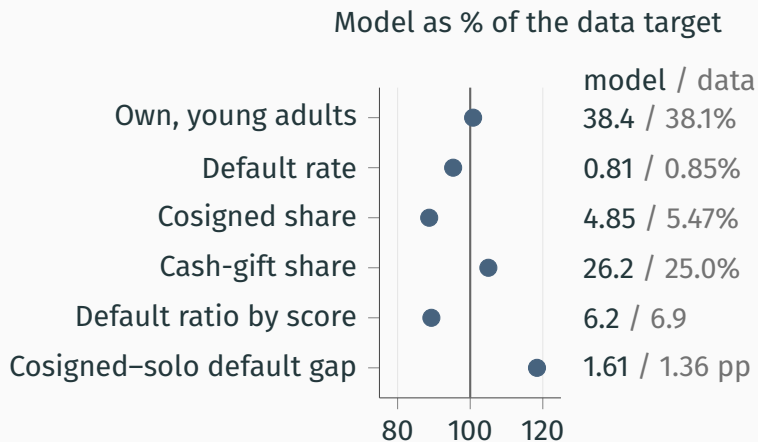
Parameter	Symbol	Value	Source
<i>Taste, and numerical settings</i>			
Parental-help dispersion	σ_{help}	0.50	structural; below 0.1 cosigning vanishes
Choice smoothing	$\sigma_{def}, \sigma_{rb}, \sigma_{menu}$	0.100	logit smoothing, not calibrated
Grid points	n_a, n_z, n_h	30, 11, 6	discretisation
Lifespan (periods)	J	70	ages 1–70

The six estimated parameters are on the previous backup: [► estimates](#)

[► back to 1/3](#)

[◄ back](#)

Backup: the targeted moments at a glance



◀ back

Backup: anatomy of the default gap

Component	pp
Across cells: where cosigners buy	+0.69
Within cells	+0.26
observable state	−0.02
hidden type	+0.28
need-based adverse selection	+1.48
parental screening	−1.20
Pooled gap, composition only	+0.95
Behaviour when the guarantee is called	+0.02

▸ after the reform

◀ back

Backup: the fair premium, and what it hangs on

Basis points per year	$\ell = 0.25$	$\ell = 0.40$
Composition charge (riskier pool)	+35.1	+56.2
Recovery credit (guarantee)	-52.2	-83.6
Actuarially fair premium	-17.1	-27.4
Premium actually charged	+0.4	
Retained by the lender	+17.5	+27.8

Sign turns on parental coverage alone: negative above $\bar{p} = 0.66$, and over **91%** of the range. Model $\bar{p} = 0.95$.

At the survey floor, $\bar{p} = 0.631$, the fair premium is +2.0 to +3.2 bp and the transfer closes.

► pp to bp

◄ back

Backup: from default rates to basis points

bp a year = default rate a year $\times$ share not repaid $\times$ loss given default ℓ

	Default rate a year	Not covered by a parent	Uncovered rate	bp a year, lender loses 25%	40%
Solo	0.81%	all	0.81%	20.2	32.3
Cosigned	2.41%	5%	0.12%	3.1	4.9
Fair premium (cosigned – solo)				–17.1	–27.4
<i>The same number, split as on the slide</i>					
Riskier pool	2.21 – 0.81 = +1.40 pp			+35.1	+56.2
Parents pay	0.12 – 2.21 = –2.09 pp			–52.2	–83.6

Parents cover 95% of what a cosigned child misses. 2.21%: the cosigned pool's default rate with no parent behind it. Rates rounded, basis points computed before rounding. Loss given default: Qi–Yang 09. The slide's bars are not these inputs: they split a gap measured on another weighting (+0.95 pp, against +1.40 here); do not multiply the bars by ℓ .

Backup: the pool the reform delivers

	Baseline	After	Δ
Need-based adverse selection (pp)	1.48	2.05	+0.57
Parental screening correction (pp)	-1.20	-1.40	-0.20
= hidden-type composition (pp)	0.28	0.65	+0.37
Invocation margin (pp)	0.02	0.29	+0.27
Safe-type share, cosigned pool	0.721	0.662	-0.059
Cosign share (%)	4.84	9.74	+4.90
House size at origination, $E[h]$	0.6061	0.6059	-0.0003

Twice the pool and riskier, and the screen grows with it. House size does not move.

Unpriced, the invocation margin rises +0.65 instead. About half the new cosigned loans replace solo loans that would have been made anyway.

Backup: the ownership gain by housing supply elasticity



Welfare: +0.237 / +0.113 / +0.074 / +0.050%

[◀ back](#)

Backup: if parents pay less, and deposit relief for everyone

Reform (changes, pp)	Owners	Missed payments	Foreclosures, parents cover 95%	63%
Cosigned 97.5%, guarantee priced	+1.27	+0.23	-0.03	+0.07
Cosigned 97.5%, not priced	+1.17	+0.25	-0.02	+0.09
Fair price only	+0.14	+0.02	-0.01	0.00
97.5% for every buyer	+9.28	+0.27	+0.19	+0.22

Fixed house prices. 63% is the survey floor for parental coverage. The 97.5%-for-all row is pending a re-run.

[◀ back](#)

Backup: why the deposit, not the price

- Price alone: rate -24 bp, cosigning $+1.20$ pp, ownership only $+0.14$ **pp**
(most of the take-up is solo borrowers switching contract)
- Deposit alone: ownership $+1.17$ **pp**, eight times the price leg
- With a parent, joint debt-to-income never exceeds **0.29** against a cap of **0.42** ► the income test is slack, the deposit is not
- λ_m^{co} from 0.90 to 0.975 cuts required assets **28 to 48%** where cosigning is common, and leaves the solo contract alone
- Can fund the deposit at $z_c = 4$: **2.5%**; at $z_c = 7$: **70%**

Backup: what survives when house prices adjust (1/2)

	Baseline	Fixed	Market clearing, by supply elasticity		
	pre-reform	prices	$\varepsilon = 4.0$	$\varepsilon = 2.2$	$\varepsilon = 1.5$
<i>Panel A. The credit market</i>					
Homeownership rate (pp)	38.40	+1.27	+0.73	+0.54	+0.43
Cosign share (pp)	4.84	+4.89	+4.64	+4.63	+4.62
Cosign income gradient	-0.012	-0.482	-0.479	-0.474	-0.470
Average cosigned rate (bp)	4.32%	-22.1	-22.3	-22.2	-22.2
Aggregate default rate (pp)	0.807	0.000	-0.004	-0.002	+0.001
Invoked guarantee (pp)	2.41	+1.34	+1.32	+1.31	+1.31
Cosigned pool risk (pp)	2.21	+0.98	+1.01	+1.01	+1.01

Baseline is the pre-reform level in each row's own units; every other column is a change. $\varepsilon = 2.2$ is the calibrated value.

► wealth, welfare, prices

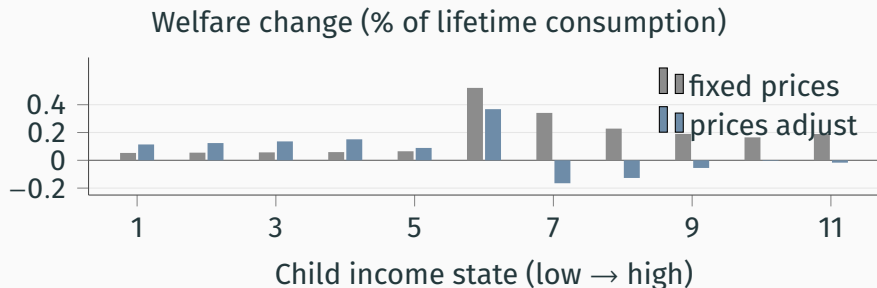
◄ back

Backup: what survives when house prices adjust (2/2)

	Baseline	Fixed	Market clearing, by supply elasticity		
	pre-reform	prices	$\varepsilon = 4.0$	$\varepsilon = 2.2$	$\varepsilon = 1.5$
<i>Panel B. Wealth, welfare and transfers</i>					
Mean wealth (%)	1.889	+2.63	+1.84	+1.62	+1.43
Wealth Gini (points)	73.19	-0.65	-0.40	-0.30	-0.24
Top-10% share (points)	43.25	-1.07	-0.65	-0.47	-0.37
Bequests (%)	6.164	+3.72	+2.80	+2.50	+2.24
Aggregate welfare, CEV (%)	—	+0.237	+0.113	+0.074	+0.050
Parent-income ownership gap	0.388	+0.574	—	+0.487	—
Parent-wealth ownership gap	0.002	+0.115	—	+0.116	—
House price, owner-occupied	12.26	—	+0.51%	+0.68%	+0.79%
Rent	1.10	—	-0.15%	-0.21%	-0.23%
Price-to-rent ratio	11.11	—	+0.66%	+0.90%	+1.02%
Rental housing stock	—	—	-0.60%	-0.47%	-0.35%

◀ back

Backup: welfare before house prices adjust



Fixed prices +0.237%, no state loses, peak +0.520 at the mode.

Clearing +0.074%: the bottom gains more, states 7 to 11 pay, at most -0.17.

▶ against the literature

◀ back

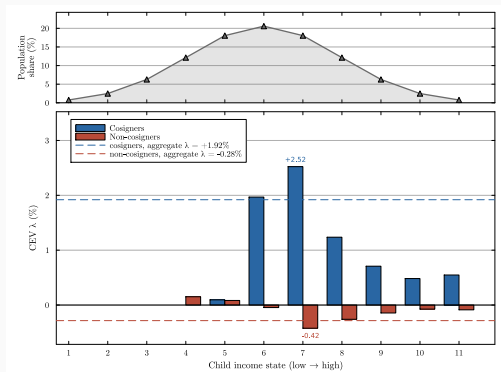
Backup: welfare against the literature

	Repricing	Reaches	CEV (%)
Jeske et al. (2013), subsidy removed	30 bp	all mortgages	+0.50
Elenev et al. (2016), GSE phase-out	255 bp	all mortgages	+0.68
leverage cap, underpricing kept	—	all mortgages	+0.11
This reform	26 bp	5–10% of loans	+0.074

Normalised by the share of the market each one reaches, the same order, and bought with a rule change rather than a transfer.

Both benchmarks are tax-financed; in Jeske et al. too, the poorer gain most. [◀ back](#)

Backup: welfare, users vs. everyone else



Cosigners +1.92%, everyone else who buys -0.28%; cosigners are 4.8% of buyers, 2.1% of the entry cohort.

Compares owning from 36 with never owning, so the users' gain is an upper bound

◀ back

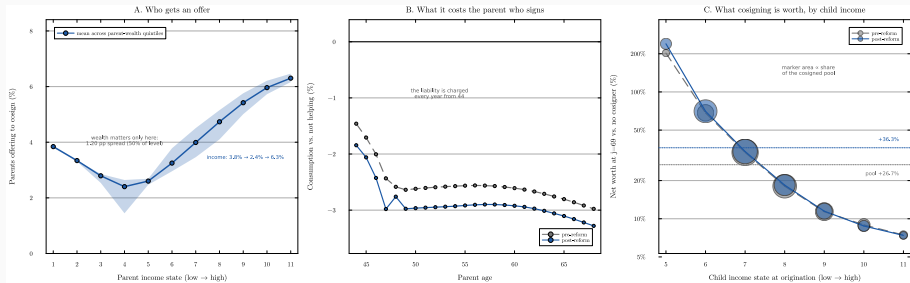
Backup: family vs. state guarantee

	Baseline	Family	State	State, net of tax
Ownership, fixed prices (%)	38.40	39.67	41.38	41.21
Guaranteed share (%)	4.8	9.7	21.3	21.3
Safe-type share, guaranteed pool	0.721	0.662	0.662	0.663
Outlay (bp of labour income)	0.79	2.64	5.90	5.85
Welfare, fixed prices (%)		+0.237	+0.374	+0.219
Welfare, prices adjust (%)		+0.074	+0.071	-0.046

Invocations beyond what the pool's risk explains rise +0.27 pp under either provider; only where the liability lands differs.

[◀ back](#)

Backup: who has a parent to ask



A. Offers peak at 6.3% at the top parent income, 3.8 at the bottom; cash gifts 22.9 → 26.6%.

B. The cosigner consumes 2.6% less every year after: 0.71 of a year's consumption, 0.80 post-reform.

C. Worth 70% more net worth at $j = 69$ at $z_c = 6$, 11% at $z_c = 9$, under 8% at $z_c = 11$.

◀ back